Adversarial Sim2Real Transfer via Cross-Domain Coupling Networks: A Single-Source Generalization Approach for Bearing Fault Diagnosis

Kun Xu, Qibin Wang, Xianguang Kong, Guowei Zhang, Ni Liu, Zhenguo Liu, and Bingquan Chen

***Abstract*—In recent years, domain generalization-based fault diagnosis methods have achieved impressive results. Most existing methods extract common diagnostic knowledge from multiple source domains and generalize it to unknown domains. However, in real industrial settings, a challenging scenario often arises where only a single domain is available for training, yet diagnosis is needed across many unknown domains. To address this issue, this study proposes a simulation-enhanced multicategory coupled generative adversarial network (MCCGAN) integrated with an adversarial strategy, designed for deployment under unknown operation conditions. Initially, a physical nonlinear vibration model of the bearing is developed to generate ample simulated fault samples, capturing the physical characteristics of faulty bearings across diverse operation conditions. Subsequently, to explore the common transformation relationship between simulated and real data, the MCCGAN architecture is meticulously designed. Leveraging the coupling and personality layers within MCCGAN, the model effectively captures both the shared and unique features between simulated and real data. Finally, an adversarial-based domain generalization network, augmented with conditional maximum mean discrepancy (MMD), is employed to further enhance model generalization capabilities and mitigate localized misclassifications. Extensive comparative experiments conducted on three datasets demonstrate that the proposed diagnostic model achieves high accuracies of 99.73%, 99.67%, and 99.82% in single-domain generalization diagnostic tasks, strongly validating the effectiveness and superiority of the proposed method.**

***Index Terms*—Generated adversarial networks, intelligent fault diagnosis, physical vibration model, simulation enhanced, single domain generalization (SDG).**

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NOMENCLATURE

MCCGAN	Multicategory coupled generative adversarial network.
CNN	Convolutional neural network.
RNN	Recurrent neural network.
DBN	Deep belief network.
SAE	Stack auto encoder.
MMD	Maximum mean discrepancy.
MAML	Model-agnostic meta-learning.
GAN	Generative adversarial network.
4-DOF	Four degrees of freedom.
IEDGNet	Intrinsic and extrinsic domain generalization network.
DGDNN	Domain generalization deep neural network.
SDG	Single domain generalization.
CoGAN	Conditional coupled GAN.
PDBN	Paderborn.
SDUST	Shandong University of Science and Technology.
$D^S = \{x_i^s, y_i^s\}_{i=1}^{n^s}$	Source domain.
$D_{1,\ldots,K}^T = \{x_i^t, y_i^t\}_{j=1}^{n^t}$	K unknown target domains.
$x_i^s \sim P(x_i^s)$	Distribution of source domain samples.
$x_j^t \sim P(x_j^t)$	Distribution of target domain samples.
$\boldsymbol{G}_k$, $\boldsymbol{D}_k$	Multi category coupled generator, multi category coupled discriminator.
$\boldsymbol{G}_1, \ldots, \boldsymbol{G}_C$	Generator with unique coupling layers and multiple personality layers.
$\boldsymbol{D}_1, \ldots, \boldsymbol{D}_C$	Discriminator with unique coupling layers and multiple personality layers.
$\theta_{\text{cop}}(\boldsymbol{G}_k), \theta_{\text{cop}}(\boldsymbol{G}_{C-k})$	Common coupling layer parameters of generator $\boldsymbol{G}_k$, $\boldsymbol{G}_{C-k}$.
$\theta_{\text{psn}}(\boldsymbol{G}_k), \theta_{\text{psn}}(\boldsymbol{G}_{C-k})$	Different personalized layer parameters of generator $\boldsymbol{G}_k$, $\boldsymbol{G}_{C-k}$.
$\theta_{\text{cop}}(\boldsymbol{D}_k), \theta_{\text{cop}}(\boldsymbol{D}_{C-k})$	Common coupling layer parameters of discriminator $\boldsymbol{D}_k$, $\boldsymbol{D}_{C-k}$.
$\theta_{\text{psn}}(\boldsymbol{D}_k), \theta_{\text{psn}}(\boldsymbol{D}_{C-k})$	Different personalized layer parameters of discriminator $\boldsymbol{D}_k$, $\boldsymbol{D}_{C-k}$.
$x_k^{S_\text{real}}, x_k^{T_G}$	Real data of the source domain, generated data of target domain.
$\boldsymbol{F}_{\text{ex}}$, $\boldsymbol{C}_c$, $\boldsymbol{D}_{\text{da}}$	Feature extractor, classifier, domain discriminator.
$L_{\text{cls}}, L_{\text{da}}, L_{\text{cmmd}}$	Classification loss, domain adversarial loss, conditional MMD loss.

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I. Introduction

With the development of modern industrialization, rotating machinery has found extensive applications across manufacturing, transportation, and aerospace fields [1]. Nevertheless, during operational processes, such equipment is prone to various faults arising from fatigue wear, external impacts, and inherent defects. Such fault occurrences can precipitate severe accidents, causing substantial economic damage and loss of life [2]. Consequently, the development of timely and reliable fault diagnosis technologies is critically important to ensure the safe and stable functioning of rotating machinery [3].

In recent years, data-driven intelligent fault diagnosis approaches have garnered substantial attention, driven by the accumulation of industrial big data and advancements in machine learning. Traditional intelligent methods predominantly emphasize on feature extraction [4] and pattern recognition [5]. In contrast, deep learning-based approaches unify these processes within an end-to-end framework, facilitating the adaptive acquisition of diagnostic knowledge directly from fault data [6]. Widely adopted deep neural network architectures, including CNNs [7], [8], RNNs [9], DBNs [10], and SAEs [11], have demonstrated notable success in the fault diagnosis of rotating machinery. These deep-learning methods primarily focus on modeling the relationship between fault patterns and features using extensive labeled in-domain datasets to achieve high diagnostic accuracy. Nevertheless, in practical operational environments, shifts in data distribution—arising from variations in equipment operating conditions, load changes, or fault progression—result in discrepancies between the feature distributions learned during training and those encountered during testing. Such distributional mismatches substantially degrade diagnostic performance and pose challenges for adapting to dynamically evolving industrial contexts.

To mitigate the challenge of distributional discrepancies frequently encountered in practical industrial environments, transfer learning has been widely employed for cross-domain fault diagnosis. These approaches generally leverage labeled data from a source domain alongside unlabeled or limited labeled data from a target domain to minimize domain divergence, thereby enhancing performance on the target domain [12]. Specifically, parameter transfer methods involve pretraining models on extensive labeled source datasets followed by fine-tuning with a small quantity of labeled target data to facilitate adaptation to the target task [13]. Metric-based methods use statistical measures, such as MMD, to reduce the distance between source and target domain distributions [14]. In addition, adversarial-based approaches apply principles from game theory to iteratively align feature representations of the source and target domains through adversarial training processes [15], [16]. These transfer learning frameworks effectively enable knowledge transfer from well-annotated source domains to target domains, addressing challenges related to limited target domain data availability and high annotation costs. Nonetheless, in scenarios where no target domain data is accessible, these methods are unable to establish meaningful correspondences between source and target domains, resulting in substantial declines in transfer efficacy and rendering them inadequate for diagnostic applications in industrial contexts characterized by scarce cross-domain data.

To address the challenge posed by the absence of target domain data and to enhance the generalization capability of diagnostic models under unknown operation conditions, fault diagnosis approaches based on multidomain generalization have progressively emerged [17]. These approaches seek to extract shared diagnostic knowledge from multiple source domains and develop domain-independent models that are both discriminative with respect to fault categories and robust against variations in operating conditions. Such methods can be broadly categorized into three groups: advanced deep architectures, metric learning, and adversarial training. For instance, Zhu et al. [18] optimized a conventional CNN by integrating a capsule network to tackle bearing domain generalization tasks. Wang et al. [19] improved domain generalization networks by incorporating center loss to learn domain-invariant features, with validation conducted on bearing datasets. Han et al. [20] employed triplet loss within a domain generalization network to enhance intraclass compactness and the interclass separability of features, thereby achieving a more generalized decision boundary. Zhang et al. [21] proposed a novel condition adversarial strategy designed to confuse the discriminator and extract generalized features, demonstrating both computational efficiency and accuracy on bearing datasets. In addition, Liao et al. [22] combined semi-supervised learning with a domain generalization network, training it on both labeled and unlabeled source data to enable the fault diagnosis model to generalize to unseen speeds. Overall, multidomain generalization-based fault diagnosis methods emphasize the collaborative learning of domain-invariant universal features from multiple related source domains to improve adaptability to target domains, facilitating cross-domain diagnosis without reliance on target domain data. Nevertheless, challenges remain in acquiring and aligning multisource domain data, and an excessive focus on domain invariance may compromise task-specific feature representation pertinent to the target domain, thereby diminishing the accuracy of detecting subtle industrial fault patterns.

Recent advancements in diagnostic methodologies have focused on single-domain generalization to tackle complex industrial challenges [23]. The objective of single-domain generalization is to develop diagnostic models capable of extracting diagnostic knowledge from a single source domain while maximizing their generalizability across multiple, previously unseen target domains. Current approaches to single-domain generalization encompass three primary techniques: meta-learning, data augmentation, and adversarial-based domain generalization. For instance, Peng et al. [24] introduced MAML alongside a domain adversarial network to explore the optimal initialization gradient, enabling rapid adaptation of the model to novel diagnostic tasks under unknown conditions. Similarly, Kutbi et al. [25] integrated a domain adaptation network with sensor fusion techniques to eliminate domain variant effects and learn common representations between source and target. Wang et al. [26] proposed a CoGAN to capture the joint distribution from relevant and irrelevant task

TABLE I
FAULT DIAGNOSIS SCENARIO SETTINGS OF DIFFERENT DIAGNOSTIC METHODS

Method	Data distribution between source and target domains	Number of source domains	Target data	Technology	Reference
Deep learning	Same	Single	No data	CNN, SAE, RNN, DBN	[4~11]
Domain adaptation	Different	Single	Unlabeled data	Finetune, MMD, Domain adversarial	[12~16]
Domain generalization	Different	Multiple	No data	advanced deep architectures, metric learning, adversarial training	[17~22]
Single domain generalization	Different	Single	No data	MAML, IEDGNet , SDG, CoGAN	[23~27]

samples in source and target domains, generating high-level concepts of nonavailable target-domain data. Zhao and Shen [27] improved the domain adversarial network with mutual information to generate pseudo target domains significantly different from the source domain, allowing the diagnosis model to learn general features resistant to unknown domain shifts. While single-domain generalization methods effectively learn domain-invariant universal features from a solitary source domain dataset—thus alleviating the challenges associated with collecting data from multiple source domains as required in multidomain generalization—they face limitations. Specifically, the universal features derived in single-domain contexts often fail to comprehensively represent the complex and variable distributions characteristic of unknown target domains, exhibit limited robustness to cross-domain discrepancies, and inadequately capture task-specific attributes pertinent to the target domain.

In light of the above analysis, Table I presents a comparative overview of various diagnostic methods across different fault diagnosis scenarios. It is evident that multidomain generalization-based diagnostic approaches demonstrate strong robustness when confronted with unknown domains; however, they necessitate a substantial volume of source domain data. Conversely, single-domain generalization-based diagnostic methods require data from only one source domain but exhibit limited adaptability to the task-specific characteristics of the target domain. Consequently, there is a pressing need to develop a fault diagnosis methodology that demands minimal source domain data for training while maintaining robust adaptability to the target domain. Simulation data generated via dynamic models encapsulate mechanistic attributes, such as component properties and operational condition features, yet discrepancies persist when compared to real-world data. GAN offers an effective means to reconcile distributional differences between heterogeneous data types. In response, this study proposes a novel simulation-enhanced MCCGAN framework. Initially, a physical vibration model of the bearing is established to generate ample simulated fault samples under varying operating conditions, investigating the variations in vibration characteristics induced by changes in operational states. Subsequently, the proposed simulation-enhanced MCCGAN is designed to learn a common transformation relationship between simulation and real data. Notably, the input of the generator is modified by incorporating physically informed simulation data instead of random noise. In addition, the architecture introduces a coupled layer and a personality layer to effectively capture the relationship between simulated and real data. Finally, an adversarial-based domain generalization network is employed to enhance generalization performance on previously unknown domains. The principal innovations and contributions can be summarized as follows.

1) This study addresses a critical industrial diagnostic challenge, specifically the diagnosis of multiple unknown target domains using data available solely from a single source domain.
2) In contrast to existing single-domain generalization diagnostic approaches that focus exclusively on domain-invariant universal features, the present work incorporates individualized diagnostic information from unknown domains by integrating simulation-enhanced data into the generative model.
3) This study introduces a simulation-enhanced MCCGAN framework capable of generating synthetic pseudo-target domain data conditioned on specified unknown domain operating conditions and fault defect magnitudes.

The remaining structure of the article is organized as follows. Section II introduces the proposed methodology, encompassing the problem definition, diagnosis framework, simulation data generated, simulation-enhanced MCCGANs, adversarial-based domain generalization network, and general training procedure. Section III provides a detailed account of two bearing experiment verifications on unknown domain diagnosis tasks. Finally, Section IV offers concluding remarks and discusses directions for future research.

A. Problem Definition

In industrial scenarios, the collection of fault data is challenging and the operating conditions exhibit substantial variability, complicating the training and evaluation of diagnostic models under uniform conditions. Furthermore, due to the real-time requirements of industrial diagnosis, target domain data is generally excluded from the training phase of diagnostic models. To address the difficulties associated with diagnostic tasks in unknown domains, a single-domain generalized diagnostic method is proposed.

Assuming that the source domain D^S consists of n^s samples with distribution $P^s(x_i^s)$ and corresponding fault labels y^s, which is formally expressed as

$$D^S = \left\{ \left(x_i^s, y_i^s\right) \middle| i = 1, 2, \ldots, n^s; x_i^s \sim P\left(x_i^s\right); y^s \in Y \right\}_{i=1}^{n^s} \quad (1)$$

where x_i^s denotes the ith sample in the source domain, and Y is the set of fault labels.

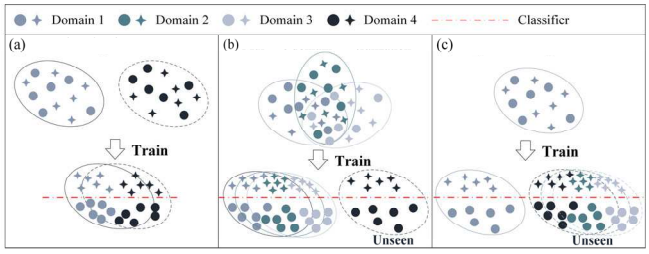

Fig. 1. Problem definition. (a) Domain adaptation. (b) Multiple domain generalization. (c) SDG.

Similarly, the unknown target domains are denoted as $\{D_k^T\}_{k=1}^K$, where represents the number of unknown target domains. Each unknown target domain contains n^t samples with distribution $P_k^t(x_j^t)$ and corresponding fault labels y_k^t, given by

$$D_k^T = \left\{\left(x_{k,j}^t, y_{k,j}^t\right) \middle| j = 1, 2, \ldots, n^t; x_j^t \sim P\left(x_j^t\right); y^t \in Y\right\}_{i=1}^{n^t} \quad (2)$$

where $x_{k,j}^t$ denotes the jth sample in the kth target domain.

As illustrated in Fig. 1, domain adaptation methods train an adaptation model by extracting diagnosis experience from both the source and target domains to achieve cross-domain diagnosis. Then, multiple domain generalization methods train the diagnosis model with multiple source domains and generalize it to the unknown target domain. Unlike these multisource domain generalization methods, SDG methods rely on training with only one source domain while striving to accurately diagnose multiple unknown target domains. It is worth mentioning that in the application of the above methods, discrepancies in sample distributions $P^s(x_i^s) \neq P_k^t(x_j^t)$ between the source and target domains arise due to the variations in operational conditions.

This article outlines the assumptions underlying single-domain generalization as follows.

1) The training procedure is confined solely to a single, known domain, with no allowance for fault data originating from multiple unknown domains.
2) The fault diagnosis task is consistent across both the source domain and the unknown target domains, implying that the same fault categories $y^s = y^t$ are present in all domains.
3) Although the source and unknown target domains share related physical characteristics, their data distributions differ $P^s(x_i^s) \neq P_k^t(x_j^t)$. In addition, the operating condition information for the unknown domain is assumed to be known.

II. Proposed Method

The diagnostic framework for the unknown domain is illustrated in Fig. 2. Specifically, a 4-DOF bearing physical vibration model is constructed to obtain simulation data across various operational conditions. Then, to generate simulation-enhanced auxiliary domain data, multicategory coupled adversarial networks are proposed. In this approach, physical information is integrated into the generated data by using simulated data as input to the generator instead

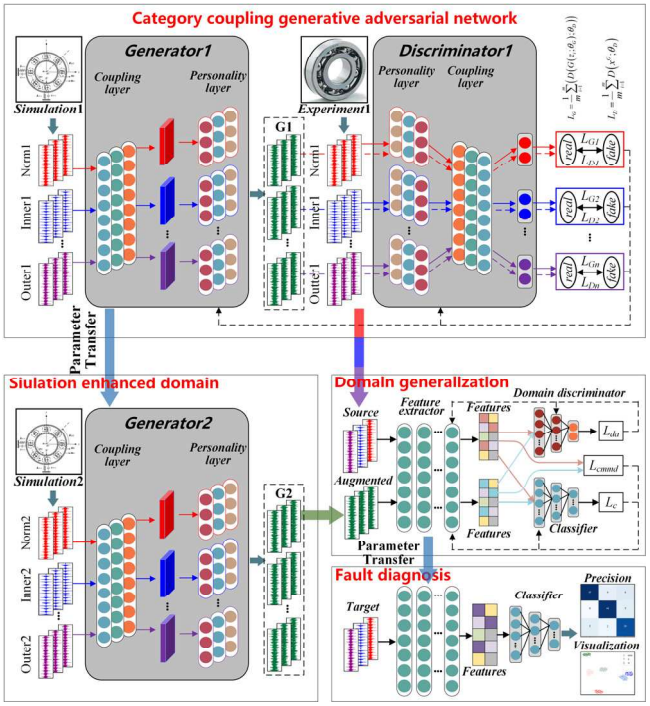

Fig. 2. Overview of the proposed diagnosis framework.

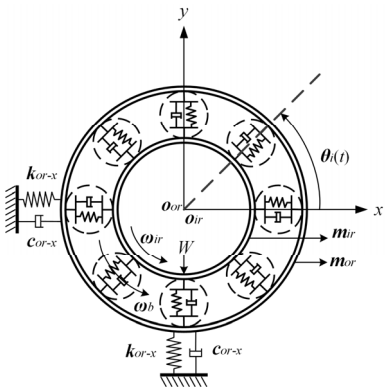

Fig. 3. 4-DOF bearing physical nonlinear vibration model.

of random noise. Furthermore, the coupling layer and personality layer are incorporated within the generator and discriminator to extract simulation-real common information and fault category individual information. Finally, an adversarial-based domain generalization network is used to learn diagnosis knowledge from both the source and the augmented domain, enabling effective generalization to the unknown target domain.

A. Simulation Data Generated by Physical Vibration Model

Owing to the universal fault mechanisms and high-fidelity vibration responses, researchers frequently employ physical vibration models to generate simulated data, thereby compensating for the scarcity of actual fault data [28]. In this study, a 4-DOF bearing physical vibration model is constructed to simulate virtual vibration signals under various operating conditions. Subsequently, a bearing dynamic model is formulated

using the lumped-parameter method derived from [29]. As shown in Fig. 3, the established bearing physical nonlinear vibration model primarily comprises the inner race (IR), roller, outer race, and foundation, with the contact interactions among these components represented by spring-mass units.

Through the analysis of the forces exerted on the bearing components during operation, the following dynamic differential equations can be derived:

$$\begin{aligned}
&m_{\mathrm{ir}}\ddot{x}_{\mathrm{ir}} + \sum_{j=1}^{N}\left(c_b\dot{\delta}_i(t) + k_b\delta_i(t)^{1.5}\right)H_i(\delta_i(t))\sin\theta_i(t) = 0\\
&m_{\mathrm{ir}}\ddot{y}_{\mathrm{ir}} + \sum_{j=1}^{N}\left(c_b\dot{\delta}_i(t) + k_b\delta_i(t)^{1.5}\right)H_i(\delta_i(t))\cos\theta_i(t) = W_y\\
&m_{\mathrm{or}}\ddot{x}_{\mathrm{or}} + c_{\mathrm{or}-x}\dot{x}_{\mathrm{or}}(t) + k_{\mathrm{or}-x}x_{\mathrm{or}}(t)\\
&\quad - \sum_{j=1}^{N}\left(c_b\dot{\delta}_i(t) + k_b\delta_i(t)^{1.5}\right)H_i(\delta_i(t))\sin\theta_i(t) = 0\\
&m_{\mathrm{or}}\ddot{y}_{\mathrm{or}} + c_{\mathrm{or}-x}\dot{y}_{\mathrm{or}}(t) + k_{\mathrm{or}-x}y_{\mathrm{or}}(t)\\
&\quad - \sum_{j=1}^{N}\left(c_b\dot{\delta}_i(t) + k_b\delta_i(t)^{1.5}\right)H_i(\delta_i(t))\cos\theta_i(t) = 0. \qquad (3)
\end{aligned}$$

Among them, m_{ir} and m_{or} denote the masses of the inner and outer races, respectively. The variables $x_{\mathrm{ir}}, y_{\mathrm{ir}}, x_{\mathrm{or}}, y_{\mathrm{or}}$ correspond to the displacements of the inner and outer races in the x- and y-directions, respectively. The parameters $c_b, k_b, c_{\mathrm{or}-x}$, and $k_{\mathrm{or}-x}$ denote the viscous damping coefficients and support stiffnesses associated with the roller and the foundation. In addition, θ_i and δ_i represent the rotational angle and the deformation of the ith roller, respectively, and $H_i(\cdot)$ denotes the Heaviside function. The vertical force W_y exerted on the IR is attributed to the weight of the shaft.

Based on the constructed bearing physical simulation model, four types of fault simulation data, including normal (N), inner race (IR) faults, outer race (OR) faults, and roller faults, can be produced across varying rotational speeds and load conditions.

B. Multi Category Coupled GANs

In data augmentation tasks, GAN-based data generation methods frequently encounter inherent limitations. One notable issue is mode collapse, characterized by the loss of category diversity and reduced variability among generated samples, which is attributed to the generator receiving input solely from a single noise distribution [30]. Consequently, it is imperative to introduce supplementary constraints aimed at mitigating mode collapse. Enhancing the generator's input by imbuing it with physically meaningful information related to the real data has been shown to improve the quality and diversity of the generated outputs [31].

While designing the input of the generator, careful attention is given to ensuring the similarity in both distribution and physical characteristics with the real data. Throughout the development of the physical vibration model, operational parameters and model parameters that align closely with those of the real data are rigorously applied. Consequently, the simulated data produced by the physical vibration model maintains a consistent fault characteristic and exhibits a data distribution comparable to that of the real data. Furthermore, for fault-simulated data across different categories, inherent differences in their distributions are preserved, thereby guaranteeing both the fidelity of generated data categories and the diversity of the samples. This approach enables the simulated data input to mitigate the risk of model collapse and to facilitate more efficient model training.

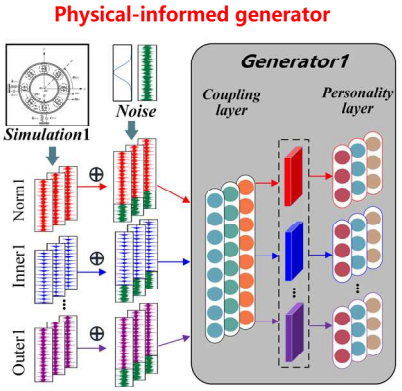

Fig. 4. Simulation enhanced generator.

The parallel concatenation of simulated data and random noise inputs facilitates the incorporation of fault mechanisms within the generated data while simultaneously promoting sample diversity. As illustrated in Fig. 4, the input of $\boldsymbol{G}_k$ comprises two branches: one representing simulation-enhanced simulated data x^{sim}, and the other consisting of random noise z. These two components are combined in parallel, which can be formally represented as follows:

$$X^{G}_{\mathrm{input}} = x^{\mathrm{sim}} \oplus z \qquad (4)$$

where X^{G}_{input} denotes the input of the generator, and x^{sim} and z, respectively, denote the simulated data and random noise. The symbol $\oplus$ denotes the concatenation operation.

Motivated by the mechanism that the identical semantic attributes are shared between simulated and real domains, a multicategory coupling layer is introduced to constrain the invariant translation relationship between simulation and real spaces. Conceptually, a transformational relationship $D^{S}_{\mathrm{real}} = \psi_{\mathrm{trans}}(D^{S}_{\mathrm{sim}})$ exists between the simulated and real domains, which can be generalized across various categories of fault samples. In particular, a weight-sharing constraint strategy is considered to ensure that high-level semantic transformation concepts remain consistent across all categories. By using a carefully designed architecture, MCCGAN effectively learns a shared distributional transformation relationship between simulated and real data.

Given the distinct semantic characteristics inherent to various fault categories, it is imperative to develop personalized layers to effectively process the low-level features associated with different fault-category samples in both simulated and real domains. In particular, a dedicated personalized layer is designed for each fault-category sample to prevent semantic interference among the individual fault categories. Through the implementation of these personalized layers, MCCGAN can

successfully capture and learn the independent transformation mappings of fault-category samples between the simulation and real domains.

As illustrated in Fig. 2, both the generator and discriminator architectures are composed of two distinct components: the coupling layer and the personality layer. During the training process, multiclass simulation data are initially fed uniformly into the coupling layer of the generator, followed by processing through the personality layer to produce generated data corresponding to different categories. In contrast, within the discriminator, the generated data and the multiclass real data are first separately input into the coupling layer, after which they are further processed through the coupling layers. The optimization objective function of MCCGAN, which comprises generator $\boldsymbol{G}_k(k = 1, 2, \ldots, C)$ and discriminator $\boldsymbol{D}_k$, can be expressed as

$$\begin{aligned}\min_{\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C}\max_{\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C} & V(\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C,\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C)\\ &= \sum_{k=1}^{C}\boldsymbol{E}_{\boldsymbol{x}_k\sim P\boldsymbol{x}_k}\left[-\log \boldsymbol{D}_k(\boldsymbol{x}_k)\right]\\ &\quad + \boldsymbol{E}_{z\sim Pz}\left[-\log\left(1-\boldsymbol{D}_{k+1}\left(\boldsymbol{G}_j(z)\right)\right)\right] \end{aligned} \tag{5}$$

where $\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C$ and $\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C$ denote the generator and discriminator, each incorporating distinct coupling layers alongside multiple personality layers. The system $\min_{\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C}\max_{\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C} V(\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C,\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C)$ comprises $\max_{\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C} V(\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C,\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C)$ and $\min_{\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C} V(\boldsymbol{G}_1,\ldots,\boldsymbol{G}_C,\boldsymbol{D}_1,\ldots,\boldsymbol{D}_C)$, which correspond to the discriminator's objective of accurately distinguishing generated data and the generator's objective of misleading the discriminator, respectively. In this context, $\sum_{k=1}^{C}$ denote the aggregate loss arising from the multiple personality layers. Furthermore, $\boldsymbol{E}_{\boldsymbol{x}_k\sim P\boldsymbol{x}_k}[-\log \boldsymbol{D}_k(\boldsymbol{x}_k)]$ and $\boldsymbol{E}_{z\sim Pz}[-\log(1-\boldsymbol{D}_{k+1}(\boldsymbol{G}_j(z)))]$ represent the expected discriminator outputs for real and generated data, respectively.

The coupling layer and the personality layer of generator $\boldsymbol{G}_k$ and discriminator $\boldsymbol{D}_k$ are governed by four specific constraints

$$\begin{aligned}\theta_{\text{cop}}(\boldsymbol{G}_k) &= \theta_{\text{cop}}(\boldsymbol{G}_{C-k})\\ \theta_{\text{psn}}(\boldsymbol{G}_k) &\neq \theta_{\text{psn}}(\boldsymbol{G}_{C-k})\\ \theta_{\text{cop}}(\boldsymbol{D}_k) &= \theta_{\text{cop}}(\boldsymbol{D}_{C-k})\\ \theta_{\text{psn}}(\boldsymbol{D}_k) &\neq \theta_{\text{psn}}(\boldsymbol{D}_{C-k})\end{aligned} \tag{6}$$

where $\theta_{\text{cop}}(\boldsymbol{G}_k)$ and $\theta_{\text{cop}}(\boldsymbol{G}_{C-k})$ denote the common coupling layer parameters of generator $\boldsymbol{G}_k$ and $\boldsymbol{G}_{C-k}$, respectively, while $\theta_{\text{psn}}(\boldsymbol{G}_k)$ and $\theta_{\text{psn}}(\boldsymbol{G}_{C-k})$ represent their distinct personalized layer parameters. Then, $C-k$ denote a difference set of C and k. Similarly,$\theta_{\text{cop}}(\boldsymbol{D}_k)$ and $\theta_{\text{cop}}(\boldsymbol{D}_{C-k})$ correspond to the common coupling layer parameters of discriminator $\boldsymbol{D}_k$ and $\boldsymbol{D}_{C-k}$, with $\theta_{\text{psn}}(\boldsymbol{D}_k)$ and $\theta_{\text{psn}}(\boldsymbol{D}_{C-k})$ denoting their respective personalized layer parameters. The first constraint mandates that the generators across multiple categories share their bottom layers, whereas the third constraint requires the discriminators to share their top layers. Conversely, the second constraint ensures that the generators maintain individualized top layers, and the fourth constraint enforces that discriminators preserve individualized lower layers. Through these combined weight-sharing and weight-individuality constraints, the MCCGAN framework effectively captures high-level, common simulation-to-real transformation concepts alongside low-level, category-specific fault transformation features. This carefully designed architecture enables MCCGAN to flexibly model both the conditional and marginal distributions inherent in simulated and real fault data.

C. Adversarial-Based Domain Generalization Network

In Section II-B, MCCGAN was introduced as a solution to the challenge of zero samples in unknown domains. Nevertheless, the diagnostic models trained solely on generated data $x_k^{T}{}_{-G}$ may exhibit suboptimal performance when applied to unknown domains. Recognizing the effectiveness of adversarial multidomain generalization network in addressing unknown domain diagnosis, this study integrates real data $x_k^{S}{}_{-\text{real}}$ from known source domains with synthetic and generated data $x_k^{T}{}_{-G}$ from unknown target domains to train the adversarial network. Furthermore, conditional MMD is introduced to reduce the intraclass variance across corresponding categories between the known source domain and the unknown target domain. Overall, this work proposes an adversarial-based domain generalization framework tailored for the task of diagnosis in unknown domains, aiming to enhance the diagnostic model's accuracy on unknown domains.

As illustrated in Fig. 5, a typical adversarial-based domain generalization network consists of feature extractor $\boldsymbol{F}_{\text{ex}}$, classifier $\boldsymbol{C}_c$ and domain discriminator $\boldsymbol{D}_{\text{da}}$. The feature extractor $\boldsymbol{F}_{\text{ex}}$ contains three convolutional layers, three pooling layers, a flatten layer, and two fully-connected layers. This structure is capable of effectively extracting high-dimensional representations $f_k^{S}{}_{-\text{real}} = \boldsymbol{F}_{\text{ex}}(x_k^{S}{}_{-\text{real}})$ and $f_k^{T}{}_{-G} = \boldsymbol{F}_{\text{ex}}(x_k^{T}{}_{-G})$ from known source real data $x_k^{S}{}_{-\text{real}}$ and unknown target generated data $x_k^{T}{}_{-G}$. Then, the classifier $\boldsymbol{C}_c$ consists of two fully-connected layers and a softmax function. Simultaneously, similar to the classifier $\boldsymbol{C}_c$, the domain discriminator also uses a two fully connected architecture but adapts with a binary cross-entropy function. In general, the domain generalization network is constructed based on a 1-D CNN, with sigmoid functions serving as the activation mechanisms.

To learn diagnostic experience from both real data $x_k^{S}{}_{-\text{real}}$ on the known source domain and generated data $x_k^{T}{}_{-G}$ on the unknown target domain, the classification loss is employed to iteratively update the model parameters of the domain generalization network. The mathematical formulation of the classification loss L_{cls} can be expressed as follows:

$$\begin{aligned}L_{\text{cls}} = &-\frac{1}{n_s}\sum_{i=1}^{n_s}\sum_{c=1}^{C}\Delta\left\{y_i^s = c\right\}\log\frac{e^{x_{c,i,k}^{s}}}{\sum_{l=1}^{C}e^{x_{c,i,l}^{s}}}\\ &-\frac{1}{n_t}\sum_{j=1}^{n_t}\sum_{c=1}^{C}\Delta\left\{y_j^t = c\right\}\log\frac{e^{x_{c,j,k}^{t}}}{\sum_{l=1}^{C}e^{x_{c,j,l}^{t}}}\end{aligned} \tag{7}$$

where n_s and n_t represent the sample sizes of $x_k^{S}{}_{-\text{real}}$ and $x_k^{T}{}_{-G}$, respectively. The symbols y_i^s and y_n^t denote the predicted labels of $x_k^{S}{}_{-\text{real}}$ and $x_k^{T}{}_{-G}$ as generated by the generalization network. The function $\Delta\{\bullet\}$ refers to a metric used to assess the accuracy of the predicted results; it returns a value of 1 if the prediction

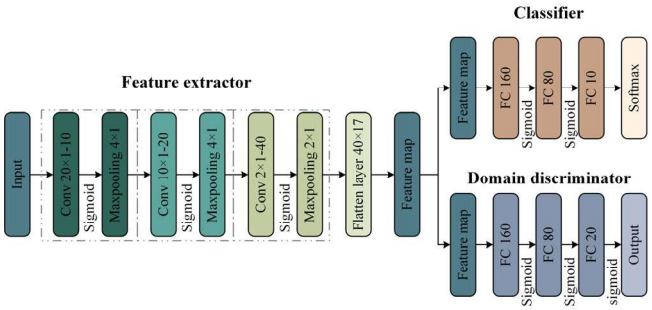

Fig. 5. Adversarial-based domain generalization network architecture.

is correct and 0 otherwise. The variable C indicates the total number of fault-categories. Furthermore, $x^s_{c,i,k}$ and $x^t_{c,j,k}$ denote the kth feature element in the output layer of classifier $\boldsymbol{C}_c$, corresponding to the ith sample in $x_k^{S_\text{real}}$ and jth sample in $x_k^{T_G}$, respectively.

To reduce the risk of overfitting and enhance the generalizability of the diagnostic model on unknown target domains, adversarial training on real data $x_k^{S_\text{real}}$ on the known source domain and generated data $x_k^{T_G}$ on the unknown domain is introduced to enforce intrinsic data generalization. Here, the binary cross-entropy loss function is adopted as adversarial loss L_{da}, which can be defined as

$$L_{\text{da}} = -\frac{1}{n_s}\sum_{i=1}^{n_s}\sum_{d=0}^{1}\boldsymbol{\Gamma}\left\{y^s_{\text{da},i}=d\right\}\log\frac{e^{x^s_{\text{da},i,d}}}{\sum_{l=1}^{1}e^{x^s_{\text{da},i,l}}} - \frac{1}{n_t}\sum_{j=1}^{n_t}\sum_{d=0}^{1}\boldsymbol{\Gamma}\left\{y^t_{\text{da},j}=d\right\}\log\frac{e^{x^t_{\text{da},j,d}}}{\sum_{l=1}^{1}e^{x^t_{\text{da},j,l}}} \tag{8}$$

where $y^s_{\text{da},i}$ and $y^t_{\text{da},j}$, respectively, denote the domain label of the source and target domains. $\boldsymbol{\Gamma}\{\bullet\}$ denote the indicator function similar to $\Delta\{\bullet\}$. $x^s_{\text{da},i,d}$ and $x^t_{\text{da},j,d}$, respectively, denote the dth feature element in the output layer of domain discriminator $\boldsymbol{D}_{\text{da}}$, taking from ith sample in $x_k^{S_\text{real}}$ and jth sample in $x_k^{T_G}$.

In addition, due to the fact that the labels of real data $x_k^{S_\text{real}}$ on the known source domain and generated data $x_k^{T_G}$ on the unknown domain have been determined, it becomes possible to align conditional distributions between $x_k^{S_\text{real}}$ and $x_k^{T_G}$. The CMMD loss L_{cmmd} is defined as

$$L_{\text{cmmd}} = \frac{1}{C}\sum_{k=1}^{C}\left\|\frac{1}{n_s}\sum\Phi\left(f_k^{S_\text{real}}\right)-\frac{1}{n_t}\sum\Phi\left(f_k^{S_\text{real}}\right)\right\|_H^2 \tag{9}$$

where $\Phi(\cdot)$ and H, respectively, denote the mapping function of Hilbert space and reproducing kernel Hilbert space.

D. Flowchart of Algorithms

With the previously described MCCGAN framework, the training procedure can be segmented into two distinct phases, each functioning within separate domains. During the initial phase, the generator $\boldsymbol{G}_k$ and discriminator $\boldsymbol{D}_k$ are optimized in the known source domain employing the Adam optimizer, which are derived as

$$\theta(\boldsymbol{G}_k) \leftarrow \theta(\boldsymbol{G}_k) + \lambda_{\boldsymbol{G}_k}\nabla_{\theta(\boldsymbol{G}_k)} \times \log\left(1-\boldsymbol{D}_k\left(\boldsymbol{G}_k\left(z;\theta(\boldsymbol{G}_k)\right);\theta(\boldsymbol{D}_k)\right)\right) \tag{10}$$

$$\theta(\boldsymbol{D}_k) \leftarrow \theta(\boldsymbol{D}_k) + \lambda_{\boldsymbol{D}_k}\nabla_{\theta(\boldsymbol{D}_k)}\log\boldsymbol{D}_k\left(x_k^{S_\text{real}};\theta(\boldsymbol{D}_k)\right) \tag{11}$$

where $\theta(\boldsymbol{G}_k)$ and $\theta(\boldsymbol{D}_k)$, respectively, denote the model parameters of generator $\boldsymbol{G}_k$ and discriminator $\boldsymbol{D}_k$. Similarly, $\lambda_{\boldsymbol{G}_k}$ and $\lambda_{\boldsymbol{D}_k}$, respectively, denote the learning rate of generator $\boldsymbol{G}_k$ and discriminator $\boldsymbol{D}_k$. Furthermore, $\nabla_{\theta(\boldsymbol{G}_k)}$ and $\nabla_{\theta(\boldsymbol{D}_k)}$, respectively, correspond to the gradients of generator $\boldsymbol{G}_k$ and discriminator $\boldsymbol{D}_k$.

The optimization objective of the adversarial-based domain generalization network includes three parts: minimizing classification loss L_{cls}, minimizing CMMD loss L_{cmmd}, and minimizing adversarial loss L_{da}. Based on traditional adversarial learning, the adversarial-based domain generalization network is optimized via (5)–(7) to obtain optimal task diagnosis modules

$$\theta(\boldsymbol{F}_{\text{ex}}) = \theta(\boldsymbol{F}_{\text{ex}}) - \eta\frac{\partial L_{\text{cls}}}{\partial\theta(\boldsymbol{F}_{\text{ex}})} - \mu\frac{\partial L_{\text{cmmd}}}{\partial\theta(\boldsymbol{F}_{\text{ex}})} - \varepsilon\frac{\partial L_{\text{da}}}{\partial\theta(\boldsymbol{F}_{\text{ex}})} \tag{12}$$

$$\theta(\boldsymbol{C}_c) = \theta(\boldsymbol{C}_c) - \eta\frac{\partial L_{\text{cls}}}{\partial\theta(\boldsymbol{C}_c)} - \mu\frac{\partial L_{\text{cmmd}}}{\partial\theta(\boldsymbol{C}_c)} \tag{13}$$

$$\theta(\boldsymbol{D}_{\text{da}}) = \theta(\boldsymbol{D}_{\text{da}}) - \varepsilon\frac{\partial L_{\text{da}}}{\partial\theta(\boldsymbol{D}_{\text{da}})} \tag{14}$$

where η, μ, and ε, respectively, denote the learning rate of classification loss L_{cls}, CMMD loss L_{cmmd}, and adversarial loss L_{da}.

The preceding sections have provided a comprehensive exposition of the specific methodologies employed within the diagnostic framework. The data flow and parameter update mechanisms of the proposed approach are depicted in Fig. 6. In addition, the step-by-step diagnosis procedures are outlined as follows.

- Step 1: A 4-DOF bearing physical nonlinear vibration model is constructed to gain supervised simulation data across various operational conditions and health states.
- Step 2: An MCCGAN is trained by the source simulation data and source real data to explore the common transformation relationship between simulation and real data.
- Step 3: The simulated data in the target domain are fed into the pretrained MCCGAN to produce the generated data that closely resemble the corresponding target real data.
- Step 4: An adversarial domain generalization model is trained using real data from the source domain alongside generated data from different target domains, thereby enhancing the diagnostic model's adaptability to diverse target operating conditions.
- Step 5: Fault diagnosis on unlabeled real data from unknown target domains is conducted using the pretrained adversarial domain generalization diagnostic model to obtain generated data-resilient diagnostic results.

III. Experimental Verification

A. Experimental Data Description

1) PDBN Dataset: The condition monitoring fault data of rolling bearings are collected from the bearing experimental platform of PDBN University [32], as shown in Fig. 7. The

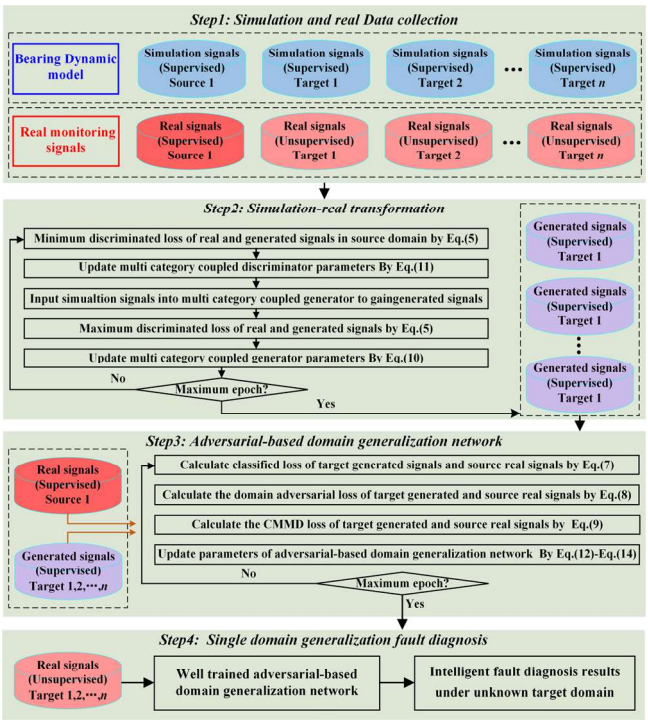

Fig. 6. Flowchart of the proposed algorithms.

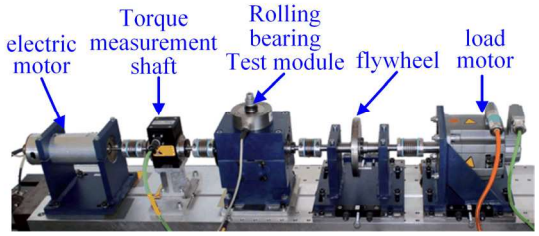

Fig. 7. Diagrams of the PDBN bearing experiment platform.

setup consists of an electric motor that drives the rolling bearing and flywheel via a shaft. Vibration signals were acquired using a piezoelectric accelerometer (PCB 336C04), characterized by a frequency range of 2–4000 Hz, a measurement range of ±500 g, and a sensitivity of 10 mV/g. This sensor was mounted on the SKF6203 bearing to capture vibrations along the x- and y-axes at a sampling frequency of 64 kHz, a rate determined in accordance with the Nyquist sampling theorem and the requirements for sampling precision. The test bearings encompass three health conditions: normal (N), OR fault, and IR fault.

The PDBN dataset consists of four operating conditions, which are established by varying the rotational speed of the drive system, the radial force applied to the bearing, and the load torque exerted on the drive system. The detailed operating parameters are listed in Table II. The test rig operates, respectively, in two rotating speeds: n = 1500 r/min and n = 900 r/min, two load torques: M = 0.7 Nm and M = 0.1 Nm, and two radial forces: F = 1000 N and F = 400 N.

TABLE II
OPERATION PARAMETERS OF PDBN EXPERIMENT PLATFORM

No.	Rotational Speed [rpm]	Load Torque [Nm]	Radial Force [N]	Name of Setting
0	1500 rpm	0.7 Nm	1000 N	N15_M07_F10
1	900 rpm	0.7 Nm	1000 N	N09_M07_F10
2	1500 rpm	0.1 Nm	1000 N	N15_M01_F10
3	1500 rpm	0.7 Nm	400 N	N15_M07_F04

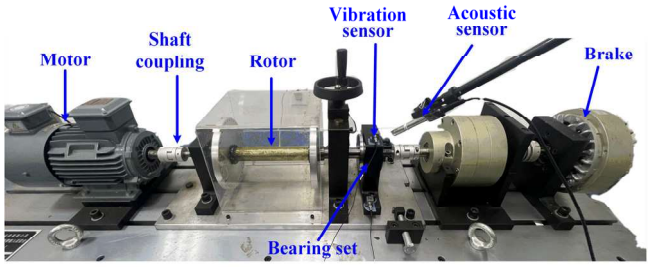

Fig. 8. Diagrams of the SDUST bearing experiment platform.

TABLE III
OPERATION PARAMETERS OF SDUST EXPERIMENT PLATFORM

No.	Rotational speed [rpm]	Radial Force [N]	Fault Size [mm]	Name of Setting
0	500 rpm	20 N	0.4 mm	N500_F20_0.4
1	500 rpm	40 N	0.4 mm	N500_F40_0.4
2	1000 rpm	20 N	0.4 mm	N1000_F20_0.4
3	1000 rpm	40 N	0.4 mm	N1000_F40_0.4
4	1500 rpm	20 N	0.4 mm	N1500_F20_0.4

2) SDUST Dataset: To overcome the limitations inherent in existing datasets and to enhance the generalizability of the proposed methodology across a variety of industrial contexts, the SDUST dataset [33] has been incorporated. This dataset originates from the bearing experimental platform at SDUST, which simulates the operational conditions of typical rotating machinery, such as fans and pumps, driven by small- to medium-sized motors, and includes representative bearing fault types. As illustrated in Fig. 8, a 1.5 kW variable frequency motor drives a rotor and bearing assembly via a shaft coupling. Vibration signals are captured using a magnetic vibration sensor (PCB 9155D), characterized by a frequency range of 0.5–7000 Hz, a measurement range of ±50 g, and a sensitivity of 10 mV/g. The sensor is mounted on the bearing assembly, and data are sampled at 25.6 kHz in both the x- and y-directions, with an overall sampling rate of 64 kHz. This configuration adheres to the Nyquist sampling theorem and meets the requirements for sampling accuracy. The test bearing is evaluated under three health conditions: normal (N), OR, and IR fault.

As illustrated in Table III, five working conditions are set on the SDUST dataset. Operating parameters include speed, radial force, and fault size. The test rig operates, respectively, in three rotating speeds: n = 500 r/min, n = 1000 r/min, and

Fig. 9. Diagrams of the BJTU bearing experiment platform.

TABLE IV
OPERATION PARAMETERS OF BJTU EXPERIMENT PLATFORM

Working condition number	Motor Speed/Load	Working condition number.	Motor Speed/Load
WC1	20Hz/0kN	WC6	60Hz/10kN
WC2	40Hz/0kN	WC7	20Hz/-10kN
WC3	60Hz/0kN	WC8	40Hz/-10kN
WC4	20Hz/10kN	WC9	60Hz/-10kN
WC5	40Hz/10kN		

n = 1500 r/min, two radial forces: F = 20 N and F = 40 N, and one fault size: L = 0.4 mm.

3) BJTU Dataset: To more accurately represent the operational working conditions of actual rail transit equipment, including variable speed, variable load, and multicomponent coupling, and to cover the common speed range of urban metro systems as well as the load range during turning, the BJTU bogie dataset is used. It is derived from a 1:2 scaled-down and simplified test rig of a real subway bogie [34] that simulates a metro bogie transmission system, as shown in Fig. 9. This test rig features a single power transmission chain consisting of an electric motor, a reduction gearbox, and axle boxes, driven by a three-phase asynchronous ac motor with speed controllable via a frequency converter. Loads are applied through a hydraulic system capable of simulating both lateral and vertical forces. The BJTU bogie dataset collects four types of signals from the test rig: three-directional vibration, three-phase current, rotational speed, and acoustic signals, with a total of 24 channels, each sampled at 64 kHz, a setting based on the Nyquist sampling theorem and requirements for sampling accuracy. The vibration signals measured by PCB 356A16, which features a frequency range of 0.5–5000 Hz, a measurement range of ±50 g, and a sensitivity of 10 mV/g. The fault scenarios include bearing faults (IR, OR, rolling element, cage), gear faults (normal, tooth root crack, tooth surface wear, tooth breakage, tooth loss), and motor faults (short circuit, broken rotor bar, shaft bending, bearing fault), covering multiple health states and operating conditions.

As illustrated in Table IV, the BJTU-RAO bogie dataset consists of nine working conditions, defined by combinations of motor rotational speed and lateral load. The test rig operates, respectively, in three rotating speeds: n = 20 Hz, n = 40 Hz, and n = 60 Hz, and two vertical forces: F = 10 kN, F = 0 kN, and F = −10 kN. As this article focuses on fault diagnosis of rolling bearings, the fault-related data of the left axle box are selected for research.

TABLE V
WORKING CONDITIONS OF UNKNOWN DOMAIN FAULT DIAGNOSIS TASKS

Dataset	Task	Source domain	Unknown target domain
PDBN	P1	N09_M07_F10	N15_M07_F10
	P2		N15_M07_F04
	P3		N15_M01_F10
	P4	N15_M07_F10	N09_M07_F10
	P5		N15_M07_F04
	P6		N15_M01_F10
SDUST	S1	N500_F20_0.4	N1000_F20_0.4
	S2		N1000_F40_0.4
	S3		N1500_F20_0.4
	S4	N1000_F20_0.4	N500_F20_0.4
	S5		N500_F40_0.4
	S6		N1500_F20_0.4
BJTU	B1	40Hz/0kN	20Hz/10kN
	B2		60Hz/-10kN
	B3	40Hz/10kN	20Hz/0kN
	B4		60Hz/-10kN
	B5	40Hz/-10kN	20Hz/0kN
	B6		60Hz/10kN

B. Unknown Domain Fault Diagnosis Tasks

In this section, 12 sets of unknown domain fault diagnosis tasks are performed on two bearing datasets to validate the generalization performance of the diagnosis framework. To cover as much as possible the range of typical operating conditions from low rotational speed and low load to high rotational speed and high load, and to systematically simulate the domain shifts that may occur in actual operation, the operating conditions are set and selected as follows. As reported in Table V, the domains of N09_M07_F10 and N15_M07_F10 in the PDBN dataset and N500_F20_0.4 and N1000_F20_0.4 in the SDUST dataset are, respectively, selected to devise SDG diagnosis tasks. Specifically, Tasks P1, P2, and P3 together constitute a single-domain generalization diagnosis task, in which the diagnostic model is trained in single source domain N09_M07_F10, but tested in multitarget domains N15_M07_F10, N15_M07_F04, and N15_M01_F10. Similarly, Tasks P4, P5, and P6 together constitute an SDG diagnosis task, in which the diagnostic model is trained in single source domain N15_M07_F10, but tested in multi target domains N09_M07_F10, N15_M07_F04, and N15_M01_F10. Tasks S1-S6 and B1-B6 are also configured according to the requirements of single-domain generalization tasks, with the only difference being the use of different datasets. In each domain of the PDBN dataset, the vibration signals of each health state is divided into 50 samples with a sample length of 2560. As for the SDUST dataset, the vibration signals of each health state is divided into 200 samples with a sample length of 1280.

C. Comparison Methods

To show the superiority of the proposed SDG diagnosis method, several state-of-the-art comparison diagnosis methods are devised. The hyperparameters of the comparison methods are set the same as those described in the literature to achieve

optimal performance. All the methods are trained with an epoch of 300 and a batch size of 50 and programmed on the PyTorch platform. Then, all methods run five times to obtain the average precision and standard deviation to eliminate the influence of random factors. In addition, to enhance the fairness of the comparison, the diagnostic network structures of all methods remain basically consistent. The detailed description of the comparison methods is implemented as follows.

1) Convolutional Neural Network: A prevalent method in the deep-learning field is selected as a benchmark method with a supervised learning scheme. In the implementation process, the diagnostic model is trained on the known source data and directly executed on the unknown target domain.

2) IEDGNet: The IEDGNet [20] combined adversarial training with triplet loss to achieve both intrinsic and extrinsic generalization. A slight variation (Gaussian noise) is applied to the source domain data to construct a novel augmented domain. Consequently, the diagnostic model is trained on known source data and an augmented domain, then executed on an unknown target domain.

3) Domain Generalization Deep Neural Network: The DGDNN [35] employed metric learning and adversarial training to extract the generalization features from different domains. A time-domain scale-stretching compression method is used to generate a novel augmented domain. Then, similar to IEDGNet, the diagnostic model is also trained on the known source domain and the augmented domain, and then executed on an unknown target domain.

4) Single Domain Generalization: The SDG method [24] introduced a Wasserstein auto-encoder to create an augmented domain and employed adversarial training to enhance the robustness of the diagnostic model. Specifically, the diagnostic model is also trained on the known source domain and the augmented domain, then executed on the unknown target domain.

5) Model-Agnostic Meta-Learning: The MAML [36] optimizes the model parameters to obtain an initialization model to adapt to different diagnostic tasks. The MAML-based diagnostic model is trained on the known source data and directly executed on the unknown target domain.

6) Conditional Coupled GAN: The CoGAN [26] captures the joint distribution from relevant and irrelevant task samples in the source and target domains, so as to generate a pseudo-target domain. Different from other comparison methods, this diagnostic model is trained on the pseudo-target domain and then applied to the unknown target domain.

7) SDCGAN: The single-domain cycle GAN [37] produces extended domains from the source domain. An adversarial contrastive training is used to learn domain-invariant generalization features from source and extended domains.

8) DEFSDG: The domain expansion fusion single-domain generalization framework [38] employed a constraint function to create and control the expanded domains from the source domain. Subsequently, a dual-branch feature fusion network is proposed to extract features, and ensemble learning and adversarial contrastive training are introduced to enhance the generalization capability of the diagnosis framework.

TABLE VI
BERING PARAMETERS OF SDUST AND PDBN PLATFORM

Bearing parameters	SKF6203 (PDBN)	SKF6205 (SDUST)
Radius of outer race	20 mm	26 mm
Radius of inner race	8.5 mm	12.5 mm
Radius of ball	3.37 mm	3.97 mm
Number of balls	8	9
Clearance of bearing	0.016 mm	0.016 mm
Stiffness coefficient of ball	3×10^8 N/m$^{1.5}$	9×10^8 N/m$^{1.5}$
Damping coefficient of ball	2000 N·s/m	5000 N·s/m
Stiffness of outer race	3×10^8 N/m$^{1.5}$	9×10^8 N/m$^{1.5}$
Damping of outer race	2000 N·s/m	5000 N·s/m

9) Proposed Method: In the training process of MCCGAN, the simulation data and real data on the known source domain are first prepared to pretrain. Then, the simulation data on the unknown target domain and the semantic constraint loss and domain extension loss are used to further finetune the MCCGAN. Next, the well-trained MCCGAN is employed to generate a pseudo-target domain. In the training process of an adversarial-based domain generalization network, the real data on the known source domain and pseudo-target domain data are employed to train the diagnostic model and tested on the unknown target domain.

D. Experimental Results and Analysis

1) Physical Vibration Model Validation: To verify the fidelity of the 4-DOF bearing physical vibration model, the periodic waveform and fault characteristic frequency of the simulation data and the real data are compared in the time domain and frequency domain. The main parameters adopted in the PDBN and SDUST experimental platform are listed in Table VI. In the PDBN experiment, the fault bearing SKF6203 is adopted with a damage diameter of 0.25 mm and a damage depth of 1 mm. And in the SDUST experiment, the fault bearing SKF6205 is adopted with a damage diameter of 0.4 mm and a damage depth of 1 mm. During the simulation process, the simulation acceleration signals of the outer race in the y-direction are collected. And the sampling frequencies of PDBN and SDUST are, respectively, 64 and 25.6 kHz.

As illustrated in Figs. 10 and 11, the time waveform and frequency spectra of simulation and real vibration signals in the PDBN and SDUST datasets are plotted. It can be observed that the vibration period and impulse characteristics of the simulated data remain consistent with the real data in the time domain. And in the frequency domain, the fault characteristic frequencies of the simulated data and the real data are still consistent. These phenomena indicate that the simulation acceleration signals generated by the 4-DOF bearing physical vibration model are able to describe the typical fault characteristics of the fault bearing under different working conditions.

The same conclusion can be drawn from the observations in Fig. 11. This result further illustrates that the physical vibration model constructed in this article can accurately simulate the fault bearings on different experimental platforms.

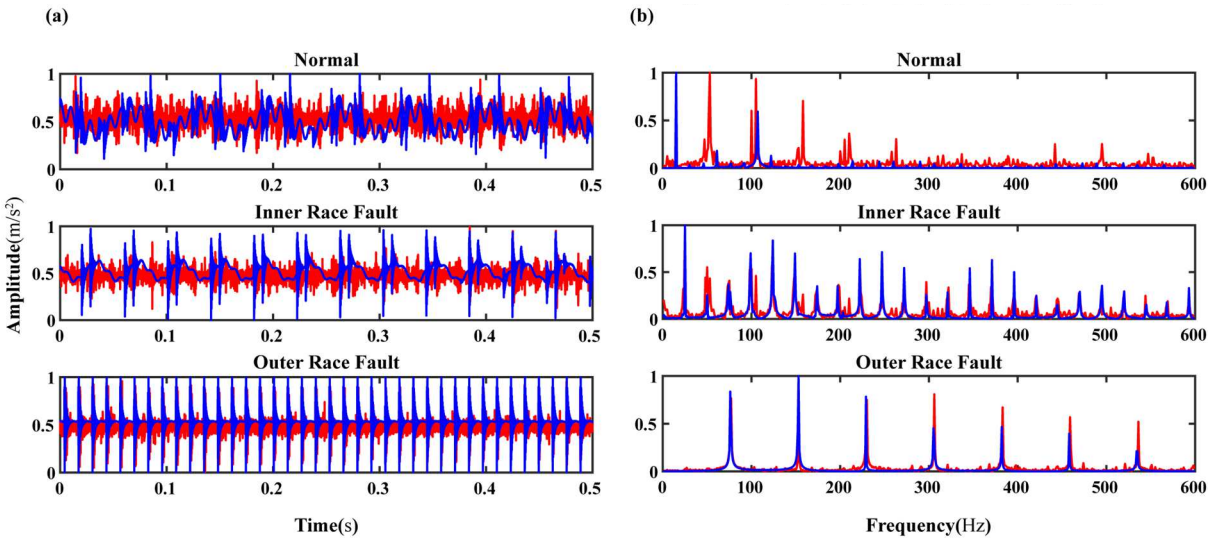

Fig. 10. Simulation and real vibration signal of PDBN on time and frequency domain. (a) Vibration signal of simulation (blue) and real (red). (b) Frequency signal of simulation (blue) and real (red).

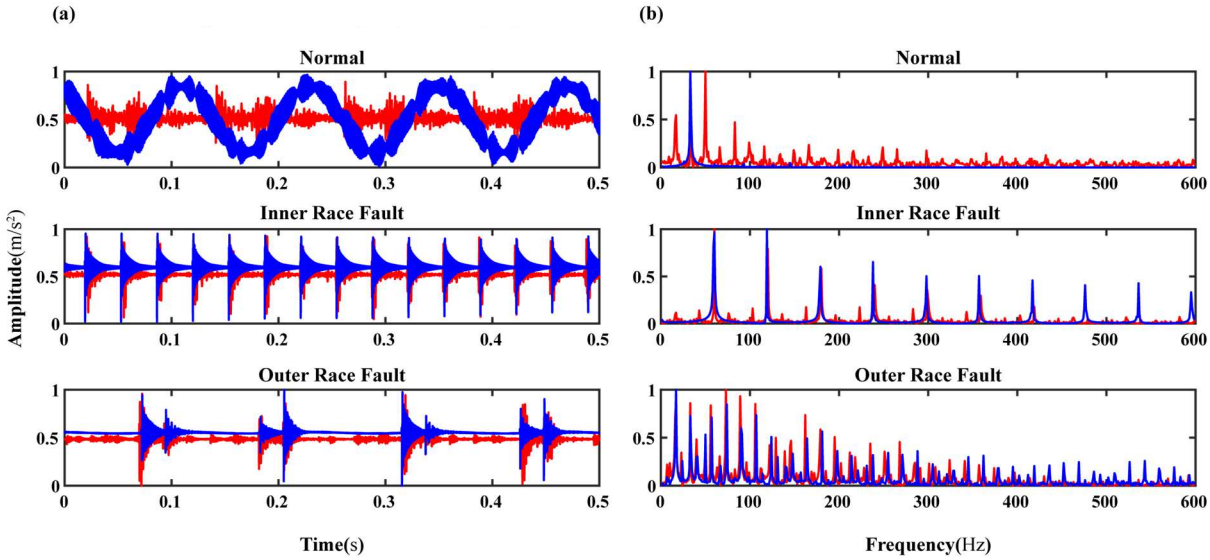

Fig. 11. Simulation and real vibration signal of SDUST on time and frequency domain. (a) Vibration signal of simulation (blue) and real (red). (b) Frequency signal of simulation (blue) and real (red).

To quantitatively evaluate the similarity between the simulated and real signals, multiple metrics are adopted, including the spectral correlation coefficient (SCC), Euclidean distance (ED), cosine similarity (CS), and Kullback–Leibler divergence (KLD), which collectively capture differences in spectral distribution, amplitude, waveform shape, and statistical fidelity.

The SCC measures the linear correlation between two signals in the time domain. It ranges from [−1, 1], where 1 indicates perfect positive correlation, 0 indicates no correlation, and −1 indicates perfect negative correlation. It is defined as (15), shown at the bottom of the next page, where $S(x_i^{\text{sim}})$ and $S(x_j^{\text{real}})$ denotes the power spectrum of x_i^{sim} and x_j^{real}.

The ED quantifies the geometric distance between two signals in vector space, reflecting their overall amplitude difference. A smaller ED indicates higher amplitude fidelity. This metric is sensitive to absolute deviations and is particularly suitable for assessing how well the simulated signal reproduces the magnitude characteristics of the real signal. It is defined as

$$d = \sqrt{\sum_{i=1}^{n} \left(x_i^{\text{sim}} - x_j^{\text{real}} \right)^2}. \tag{16}$$

The CS measures the cosine of the angle between two signal vectors, reflecting the similarity in their waveform patterns, independent of amplitude scaling. The value ranges from [−1, 1], with values closer to 1 indicating greater similarity in shape. CS is especially useful for evaluating pattern consistency, such as whether fault-induced impulses and their temporal structures are accurately reproduced in the simulation. It is widely used in waveform matching and signal

TABLE VII
QUANTITATIVE COMPARISON RESULTS BETWEEN SIMULATED AND REAL SIGNALS

Dataset	Health state	SCC	ED	CS	KL
PDBN	Normal	0.186	11.783	0.030	2.947
	Inner race fault	0.772	6.539	0.225	1.842
	Outer race fault	0.780	7.504	0.206	2.099
SDUST	Normal	0.635	3.845	0.084	1.766
	Inner race fault	0.854	3.581	0.275	1.684
	Outer race fault	0.932	3.573	0.445	1.018

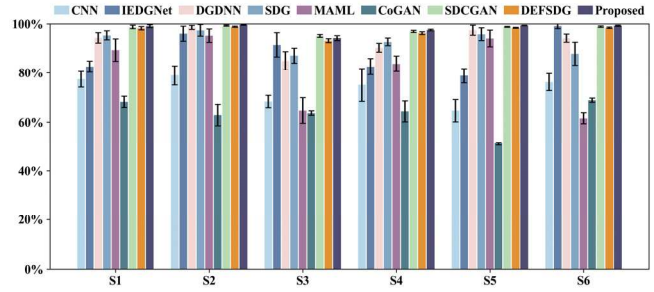

Fig. 12. Average accuracies of comparative methods on PDBN.

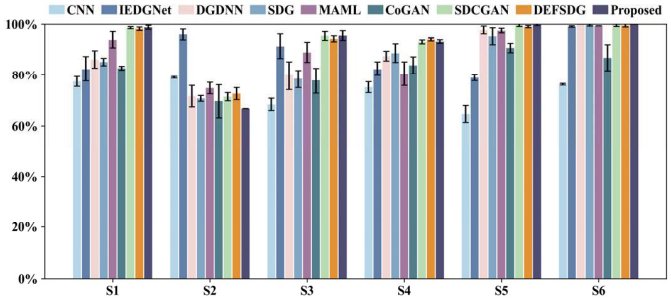

Fig. 13. Average accuracies of comparative methods on SDUST.

shape analysis. It is defined as

$$\cos = \frac{\sum_{i=1}^{n} x_i^{\text{sim}} \cdot x_j^{\text{real}}}{\sqrt{\sum_{i=1}^{n} \left(x_i^{\text{sim}}\right)^2} \cdot \sqrt{\sum_{i=1}^{n} \left(x_j^{\text{real}}\right)^2}}. \tag{17}$$

The KLD measures the information discrepancy between two probability distributions and is commonly used for comparing spectral or statistical characteristics. For two discrete distributions P and Q, representing the spectra of the real and simulated signals, respectively, it is defined as

$$D_{\text{KL}}(P \| Q) = \sum_{i=1}^{n} P\left(x_i^{\text{sim}}\right) \cdot \log\left(\frac{P\left(x_i^{\text{sim}}\right)}{Q\left(x_j^{\text{real}}\right)}\right). \tag{18}$$

As shown in Table VII, the quantitative results on the PDBN dataset show lower similarity in normal conditions,

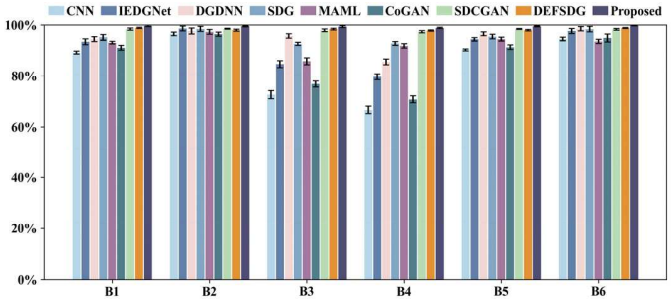

Fig. 14. Average accuracies of comparative methods on BJTU.

with SCC at 0.186 and CS at 0.030, ED reaching 11.783, and KLD as high as 2.947, indicating limited fidelity in capturing healthy dynamics. However, performance improves under fault conditions, where SCC increases to over 0.77, KLD drops below 2.1, and CS rises, suggesting effective reproduction of fault-related features. In contrast, the SDUST dataset shows consistently higher similarity across all states, with SCC reaching up to 0.932, ED around 3.6, and KLD as low as 1.018, demonstrating superior signal fidelity, especially for fault cases.

2) Unknown Domain Diagnosis Task Validation: The diagnosis results of seven comparative methods on twelve unknown domain diagnosis tasks are illustrated in Tables VIII–X, Fig. 12 (DPBN), Fig. 13 (SDUST), and Fig. 14 (BJTU). The table provides precise numerical data, allowing for clear reference to specific results across scenarios and methods. The bar chart offers visual clarity on comparative trends, making performance differences between methods intuitively understandable. To eliminate the influence of random factors, each comparative method is run five times, and the mean and standard deviation are taken.

Here, Accuracy, Precision, Recall, and $F1$-score are adopted as evaluation metrics. Among them, accuracy refers to the proportion of correctly predicted samples by the model to the total number of samples, and its calculation formula is

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{19}$$

Precision refers to the proportion of truly positive examples among the samples predicted as positive by the model. It measures the accuracy of the model in predicting positive examples, and its calculation formula is

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{20}$$

Recall, also known as completeness, refers to the proportion of truly positive examples that are correctly predicted as

$$\rho_{x_i^{\text{sim}}, x_j^{\text{real}}} = \frac{\sum_{i=1}^{n} \left(S\left(x_i^{\text{sim}}\right) - \bar{S}\left(x_i^{\text{sim}}\right)\right) \cdot \left(S\left(x_j^{\text{real}}\right) - \bar{S}\left(x_j^{\text{real}}\right)\right)}{\sqrt{\sum_{i=1}^{n} \left(S\left(x_i^{\text{sim}}\right) - \bar{S}\left(x_i^{\text{sim}}\right)\right)^2} \cdot \sqrt{\sum_{i=1}^{n} \left(S\left(x_j^{\text{real}}\right) - \bar{S}\left(x_j^{\text{real}}\right)\right)^2}} \tag{15}$$

TABLE VIII
AVERAGE TESTING ACCURACIES OF COMPARATIVE METHODS ON PDBN

Task	Metric	CNN	IEDGNet	DGDNN	SDG	MAML	CoGAN	SDCGAN	DEFSDG	Proposed
P1	*Acc*	75.86±1.96	84.26±4.93	86.27±3.39	85.33±1.46	94.00±3.18	82.87±0.75	98.67±0.44	98.21±0.68	**98.80±0.78**
	Precision	75.78±1.75	84.18±3.98	86.17±3.18	85.29±1.15	93.98±2.65	82.92±0.98	98.72±0.57	98.16±0.77	**98.94±0.81**
	Recall	75.45±2.28	84.48±4.54	86.08±2.59	85.47±2.08	93.87±1.96	82.71±1.14	98.77±0.67	98.24±0.62	**98.71±0.86**
	F1 Score	75.23±1.59	83.94±3.38	85.97±2.91	84.85±1.82	93.81±3.55	82.15±1.08	98.23±0.71	97.88±0.75	**98.46±0.95**
P2	*Acc*	64.27±0.33	68.80±2.08	71.46±4.20	70.60±1.13	**74.67±2.26**	69.60±6.43	71.26±1.58	72.45±2.36	66.67±0.00
	Precision	64.08±0.49	68.95±1.58	71.54±3.95	70.60±1.13	**74.57±2.12**	69.60±6.43	71.35±1.49	72.39±1.99	44.44±0.00
	Recall	64.33±0.56	68.88±2.11	71.38±4.33	70.60±1.13	**74.72±2.09**	69.60±6.43	71.19±1.74	72.49±2.33	66.67±0.00
	F1 Score	63.95±0.84	68.41±1.86	71.09±3.42	70.60±1.13	**74.04±1.97**	69.60±6.43	70.84±1.67	72.12±2.51	53.33±0.00
P3	*Acc*	68.53±2.44	74.13±4.76	79.73±5.59	78.40±3.54	89.07±3.92	77.73±5.04	95.49±1.69	94.32±1.16	**95.60±1.82**
	Precision	68.39±2.24	74.14±4.24	79.81±5.19	78.35±3.69	88.98±3.75	77.65±5.22	95.48±1.82	94.35±1.21	**95.57±1.66**
	Recall	68.49±2.18	74.07±3.94	79.69±5.27	78.29±3.77	89.14±4.06	77.78±5.17	95.42±1.76	94.29±1.08	**95.49±1.94**
	F1 Score	68.16±2.52	73.97±4.49	79.27±4.94	77.98±3.86	88.65±3.49	77.53±4.92	95.08±1.84	94.01±1.22	**95.04±1.67**
P4	*Acc*	75.93±2.11	80.13±2.87	87.60±1.91	88.80±3.61	80.53±4.87	84.00±3.32	93.19±0.78	**94.17±0.58**	93.33±0.73
	Precision	75.89±2.37	80.09±2.79	87.56±1.75	88.75±3.18	80.49±4.59	83.94±3.57	93.08±0.74	**94.15±0.71**	93.46±0.89
	Recall	75.91±2.49	80.16±3.05	87.71±1.86	88.84±3.75	80.38±4.66	84.12±3.23	93.13±0.67	**94.07±0.49**	93.18±0.58
	F1 Score	75.69±1.86	79.96±2.94	87.15±20.8	88.28±3.48	80.14±5.16	83.59±3.84	92.89±0.54	**93.53±0.66**	93.01±0.49
P5	*Acc*	87.73±3.23	95.34±1.12	97.74±1.55	95.33±3.27	97.47±0.89	90.80±1.86	99.61±0.51	98.98±0.42	**99.73±0.33**
	Precision	87.59±3.58	95.27±1.29	97.89±1.76	95.17±3.54	97.54±1.09	90.94±1.95	99.67±0.49	98.91±0.36	**99.67±0.28**
	Recall	87.38±3.46	95.41±1.38	97.58±1.29	95.42±3.11	97.32±0.94	90.75±1.49	99.58±0.36	98.89±0.49	**99.85±0.58**
	F1 Score	87.25±3.76	94.76±1.59	97.06±1.84	94.88±2.85	97.03±1.11	90.07±1.71	99.15±0.47	98.65±0.51	**99.24±0.46**
P6	*Acc*	99.07±0.32	99.87±0.27	**100.00±0.0**	99.60±0.33	99.33±0.00	86.93±5.10	99.84±0.76	99.42±0.58	**100.00±0.0**
	Precision	99.15±0.38	99.59±0.36	**100.00±0.0**	99.54±0.56	99.49±0.00	86.77±5.45	99.78±0.59	99.38±0.49	**100.00±0.0**
	Recall	99.21±0.54	99.77±0.59	**100.00±0.0**	99.74±0.29	99.24±0.00	86.81±4.78	99.87±0.42	99.45±0.39	**100.00±0.0**
	F1 Score	98.49±0.47	99.19±0.38	**100.00±0.0**	99.08±0.42	98.75±0.00	86.48±5.57	99.57±0.51	99.21±0.61	**100.00±0.0**

TABLE IX
AVERAGE TESTING ACCURACIES (%) OF COMPARATIVE METHODS ON SDUST

Task	Metric	CNN	IEDGNet	DGDNN	SDG	MAML	CoGAN	SDCGAN	DEFSDG	Proposed
S1	*Acc*	77.27±3.16	82.47±2.35	94.30±2.12	95.33±1.87	89.30±4.57	68.13±2.25	98.72±0.69	98.23±0.66	**99.00±0.57**
	Precision	77.19±3.28	82.51±2.44	94.18±2.28	95.14±1.56	89.44±4.13	68.01±2.33	98.71±0.75	98.25±0.79	**98.91±0.67**
	Recall	77.34±3.32	82.32±2.14	94.35±2.08	95.45±1.92	89.17±4.41	68.26±2.07	98.75±0.51	98.19±0.75	**98.77±0.49**
	F1 Score	76.88±2.84	82.01±2.68	93.86±2.24	94.79±1.58	88.86±4.12	67.74±2.59	98.27±0.65	97.89±0.84	**98.54±0.78**
S2	*Acc*	78.90±3.92	96.00±3.05	98.57±0.82	97.37±2.35	95.20±2.72	62.80±4.33	99.29±0.21	98.82±0.25	**99.60±0.14**
	Precision	78.94±4.17	95.91±2.94	98.61±0.77	97.29±2.21	95.08±2.84	62.91±4.56	99.31±0.31	98.87±0.19	**99.67±0.19**
	Recall	78.82±3.56	96.08±3.17	98.68±0.96	97.42±2.44	95.34±2.54	62.75±4.04	99.27±0.19	98.80±0.34	**99.56±0.28**
	F1 Score	78.31±3.16	95.82±2.87	98.04±0.94	96.06±2.04	94.72±2.15	62.18±4.56	98.89±0.25	98.38±0.28	**99.07±0.31**
S3	*Acc*	68.27±2.50	91.43±4.95	81.87±3.81	87.03±3.07	64.67±5.18	63.63±0.92	**95.08±0.58**	93.15±0.84	94.20±0.95
	Precision	68.34±2.67	91.49±5.02	81.81±3.92	87.15±3.21	64.34±5.21	63.59±1.09	**94.92±0.53**	93.17±0.86	94.17±0.86
	Recall	68.14±2.38	91.37±4.66	81.94±3.77	87.11±3.14	64.15±4.92	63.72±1.14	**95.01±0.74**	94.25±1.06	94.25±1.06
	F1 Score	67.95±2.07	91.01±4.34	81.24±3.75	86.54±2.86	64.24±5.03	63.13±0.74	**94.64±0.69**	93.81±0.98	93.81±0.98
S4	*Acc*	75.00±6.63	82.53±3.29	90.30±1.81	92.67±1.53	83.67±3.19	64.31±4.17	96.97±0.44	96.23±0.57	**97.50±0.32**
	Precision	74.92±6.85	82.62±3.13	90.08±1.74	92.48±1.39	83.56±3.05	64.38±4.46	96.89±0.35	96.15±0.37	**97.15±0.29**
	Recall	75.04±5.95	82.48±3.31	90.45±1.96	92.74±1.82	83.77±3.36	64.18±4.24	96.91±0.51	96.21±0.42	**97.34±0.48**
	F1 Score	74.68±6.13	82.09±2.91	89.88±1.59	92.14±1.34	83.15±2.94	63.97±4.23	96.67±0.39	95.77±0.52	**97.03±0.46**
S5	*Acc*	64.60±4.50	78.74±2.97	97.43±2.02	95.80±2.63	94.07±3.41	51.17±0.62	98.75±0.18	98.41±0.23	**99.37±0.12**
	Precision	64.72±4.15	78.88±3.18	97.18±1.91	95.87±2.43	94.15±3.28	51.29±0.77	98.89±0.31	98.37±0.24	**99.39±0.19**
	Recall	64.78±4.24	78.82±3.06	97.56±2.11	95.63±2.52	94.18±3.56	51.34±0.49	98.82±0.21	98.44±0.25	**99.42±0.15**
	F1 Score	64.23±4.15	78.16±2.87	97.08±1.95	95.26±2.11	93.56±3.08	50.69±0.51	98.46±0.37	98.05±0.31	**99.02±0.21**
S6	*Acc*	76.13±3.47	99.10±1.04	94.17±1.67	87.80±4.74	61.53±2.22	68.80±0.87	98.90±0.31	98.37±0.28	**99.24±0.23**
	Precision	76.24±3.58	99.24±0.89	94.18±1.43	87.86±4.56	61.47±2.19	68.68±0.56	98.68±0.26	98.40±0.31	**99.32±0.38**
	Recall	76.21±3.26	99.16±0.95	94.24±1.53	87.79±4.38	61.56±2.31	68.96±0.95	98.77±0.44	98.42±0.27	**99.28±0.31**
	F1 Score	75.67±3.04	98.68±1.12	93.86±1.42	87.25±4.12	61.18±2.06	68.19±0.79	98.51±0.29	98.04±0.29	**98.85±0.34**

positive by the model. It measures the model's ability to capture positive examples, and its calculation formula is

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{21}$$

The $F1$-score is the harmonic mean of Precision and Recall. It integrates both Precision and Recall and can evaluate the model performance more comprehensively. The $F1$-score will be high only when both Precision and Recall are high, and its calculation formula is

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{22}$$

where TP, FP, TN, FN represent true positive, false positive, true negative, false negative, respectively.

Experimental results across the PDBN, SDUST, and BJTU datasets consistently demonstrate the proposed method's

TABLE X
AVERAGE TESTING ACCURACIES (%) OF COMPARATIVE METHODS ON BJTU

Task	Metric	CNN	IEDGNet	DGDNN	SDG	MAML	CoGAN	SDCGAN	DEFSDG	Proposed
B1	*Acc*	89.20±0.56	93.45±1.06	94.45±0.90	95.15±1.09	93.14±0.54	91.14±0.94	98.34±0.44	98.83±0.26	**99.63±0.29**
	Precision	89.14±0.69	93.39±1.04	94.54±0.81	95.21±1.15	93.23±0.60	91.20±0.89	98.04±0.36	98.47±0.30	**99.54±0.31**
	Recall	89.23±0.75	93.35±1.27	94.41±1.05	95.26±0.96	93.18±0.49	91.23±0.81	98.20±0.40	98.75±0.33	**99.70±0.34**
	F1 Score	88.59±0.64	92.86±0.94	93.87±0.81	94.47±1.04	92.68±0.41	90.68±0.80	97.69±0.40	98.18±0.24	**99.19±0.25**
B2	*Acc*	96.50±0.72	98.58±0.94	97.62±1.17	98.42±0.98	97.27±0.98	96.38±0.83	98.49±0.22	97.94±0.38	**99.50±0.23**
	Precision	96.43±0.81	98.49±1.02	97.38±1.08	98.08±0.89	97.20±0.67	96.35±0.76	98.44±0.25	97.95±0.30	**99.45±0.26**
	Recall	96.55±0.78	98.68±0.99	97.73±1.25	98.40±1.06	97.25±1.04	96.47±1.02	97.51±0.32	97.02±0.47	**98.52±0.33**
	F1 Score	96.04±0.59	98.17±0.78	96.94±0.83	98.02±0.95	96.94±0.80	95.99±0.92	97.96±0.31	97.47±0.46	**98.97±0.32**
B3	*Acc*	72.57±1.59	84.65±1.34	95.68±0.86	92.68±0.64	85.80±1.35	76.80±1.15	97.87±0.51	98.36±0.35	**99.37±0.36**
	Precision	72.57±1.59	84.48±1.19	95.71±0.77	92.45±0.52	85.37±1.22	76.37±1.12	97.89±0.57	98.38±0.41	**99.39±0.42**
	Recall	72.57±1.59	84.71±1.40	95.59±0.72	92.71±0.68	85.86±1.42	76.86±1.22	97.93±0.47	98.42±0.31	**99.43±0.32**
	F1 Score	71.57±1.59	83.68±1.12	94.68±0.65	91.69±0.47	84.56±1.18	75.56±0.88	97.34±0.56	97.83±0.40	**98.84±0.41**
B4	*Acc*	66.57±1.50	79.70±1.12	85.55±1.15	92.82±0.73	91.87±0.85	70.72±1.40	97.33±0.39	97.82±0.23	**98.83±0.24**
	Precision	66.64±1.94	79.49±1.34	85.50±1.06	92.74±0.46	91.62±0.79	70.45±1.43	97.26±0.45	97.75±0.29	**98.76±0.30**
	Recall	66.84±1.89	79.92±1.54	85.63±1.21	92.79±0.83	91.97±0.92	70.84±1.29	97.28±0.37	97.77±0.21	**98.78±0.22**
	F1 Score	65.81±2.14	79.21±1.49	84.98±1.24	92.33±0.69	91.25±0.78	69.98±1.09	96.98±0.39	97.47±0.23	**98.48±0.24**
B5	*Acc*	90.27±0.36	94.40±0.61	96.50±0.76	95.42±0.90	94.47±0.68	91.35±0.89	98.42±0.16	97.93±0.32	**99.43±0.17**
	Precision	90.38±0.46	94.46±0.58	96.36±0.70	95.28±0.78	94.36±0.73	91.29±0.77	98.40±0.24	97.91±0.40	**99.41±0.25**
	Recall	90.46±0.58	94.57±0.88	96.68±0.69	95.65±0.87	94.53±0.69	91.42±0.88	98.47±0.14	97.98±0.30	**99.48±0.15**
	F1 Score	89.65±0.41	93.93±0.48	95.98±0.87	94.89±0.99	94.02±0.59	90.85±0.58	98.06±0.20	97.57±0.36	**99.07±0.21**
B6	*Acc*	94.53±0.69	97.63±0.99	98.50±0.85	98.35±1.05	93.62±0.75	94.90±1.50	98.30±0.34	98.80±0.18	**99.80±0.19**
	Precision	94.61±0.55	97.51±0.86	98.39±0.68	98.29±1.03	93.34±0.58	94.84±1.18	98.34±0.37	98.83±0.21	**99.84±0.22**
	Recall	94.49±0.74	97.54±0.87	98.47±0.78	98.37±0.92	93.51±0.69	94.86±1.46	98.32±0.39	98.81±0.23	**99.82±0.24**
	F1 Score	94.03±0.87	96.95±1.01	97.88±0.89	97.85±0.94	93.18±0.54	94.33±1.09	97.95±0.36	98.44±0.20	**99.45±0.21**

superiority. On PDBN, baseline CNN (75.86%) highlights the limitations of direct deep learning in unknown domains, while data augmentation-domain generalization methods (IEDGNet: 84.26%; DGDNN: 86.27%; SDG: 85.33%) and pure data generation (CoGAN: 82.87%) show moderate performance but lag behind our approach. Meta-learning (MAML: 94.00%) and adversarial learning-based methods (SDCGAN: 98.67%; DEFSDG: 98.21%) perform well, yet our method achieves the highest accuracy (98.80%).

This trend holds on SDUST and BJTU, confirming that our method, which uses simulation-enhanced multicategory coupled GANs and domain generalization networks, outperforms comparative approaches in generalization and robustness. Its consistent superiority across datasets validates strong potential for practical unknown-domain diagnosis tasks, particularly in industrial scenarios like high-speed train bogie fault diagnosis.

To intuitively exhibit the diagnostic effectiveness of the proposed method on each health states, the diagnosis results of comparative methods are demonstrated by confusion matrix, as shown in Figs. 15–17.

On the PDBN dataset, CNN, IEDGNet, and SDG exhibit high misdiagnosis rates for IR faults. DGDNN shows slight confusion between inner and OR faults, while MAML has mild misdiagnoses for normal and IR faults, which indicates that the augmented domains generated by MAML and DGDNN are more faithful than those from IEDGNet and SDG. CoGAN, a pure data augmentation method, misdiagnoses all health states, suggesting its generated data deviates from the real data distribution in the unknown target domain. Adversarial learning-based methods SDCGAN and DEFSDG perform comparably to our method with minimal misclassifications, and our method excels across all states, demonstrating strong data augmentation performance in unknown domains.

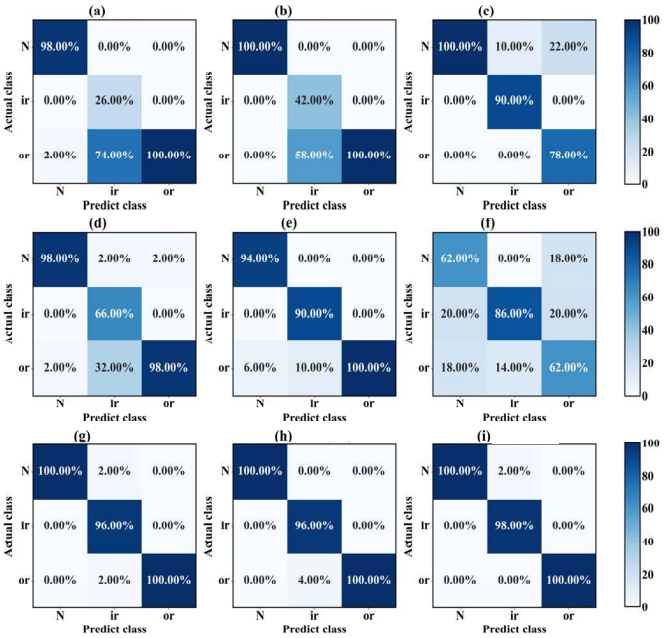

Fig. 15. Confusion matrix of comparative methods on P1. (a) CNN. (b) IEDGNet. (c) DGDNN. (d) SDG. (e) MAML. (f) CoGAN. (g) SDCGAN. (h) DEFSDG. (i) Proposed.

Similar trends are observed on the SDUST dataset: CNN, IEDGNet, MAML, and CoGAN misdiagnose normal and OR faults notably; DGDNN performs slightly weaker across the three states; SDG shows distinct confusion in OR faults. SDCGAN, DEFSDG, and our method exhibit almost no misclassifications, validating our method's strong universality and high accuracy.

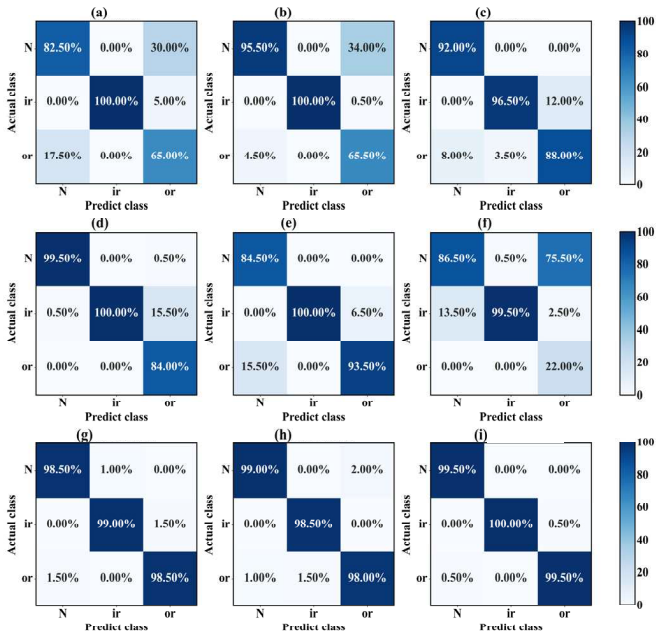

Fig. 16. Confusion matrix of comparative methods on S1. (a) CNN. (b) IEDGNet. (c) DGDNN. (d) SDG. (e) MAML. (f) CoGAN. (g) SDCGAN. (h) DEFSDG. (i) Proposed.

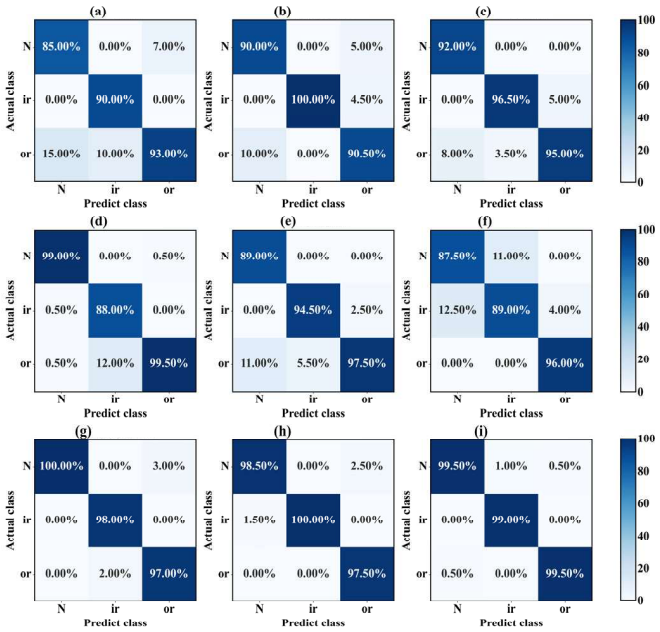

Fig. 17. Confusion matrix of comparative methods on B1. (a) CNN. (b) IEDGNet. (c) DGDNN. (d) SDG. (e) MAML. (f) CoGAN. (g) SDCGAN. (h) DEFSDG. (i) Proposed.

On the BJTU dataset, CNN, IEDGNet, MAML, and CoGAN misdiagnose OR faults notably. DGDNN performs weakly across all states, while SDG shows significant confusion in identifying OR faults. SDCGAN, DEFSDG, and our method remain nearly error-free. These consistent results across datasets confirm our method's robust universality and high diagnostic accuracy, even in complex real-world scenarios like the BJTU dataset.

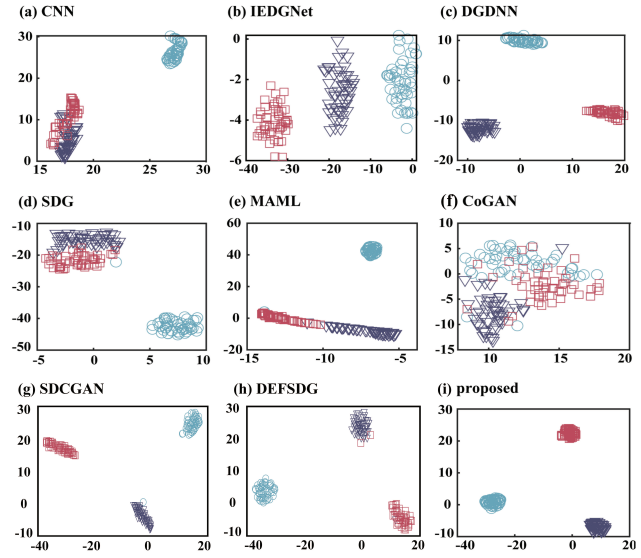

Fig. 18. Feature visualization of comparative methods on P1. (a) CNN. (b) IEDGNet. (c) DGDNN. (d) SDG. (e) MAML. (f) CoGAN. (g) SDCGAN. (h) DEFSDG. (i) Proposed.

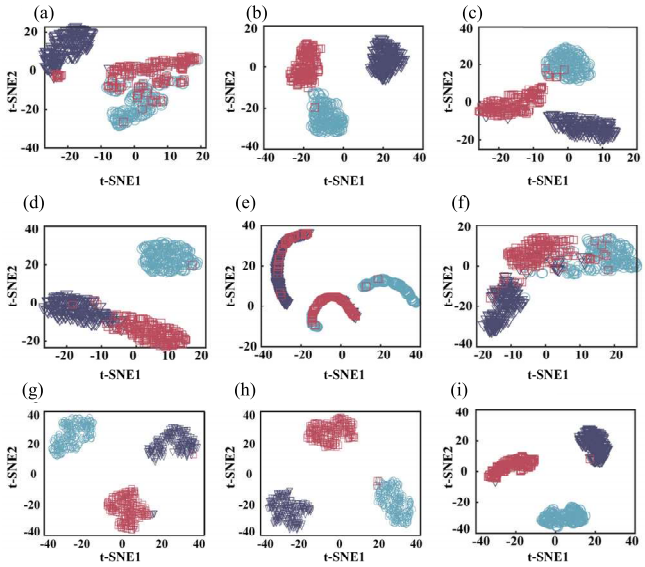

Fig. 19. Feature visualization of comparative methods on S1. (a) CNN. (b) IEDGNet. (c) DGDNN. (d) SDG. (e) MAML. (f) CoGAN. (g) SDCGAN. (h) DEFSDG. (i) Proposed.

Across the PDBN, SDUST, and BJTU datasets, a consistent pattern holds: comparative methods show varying misdiagnosis, from high rates in specific faults such as inner and OR faults to broader confusion across health states, revealing limitations in handling unknown domains. In contrast, the proposed method alongside SDCGAN and DEFSDG consistently achieves near-error-free performance, with our approach standing out for robust universality. These results validate that the proposed method delivers high diagnostic accuracy across diverse scenarios, including complex real-world settings like the BJTU dataset, underscoring its strong applicability to unknown domain diagnosis tasks.

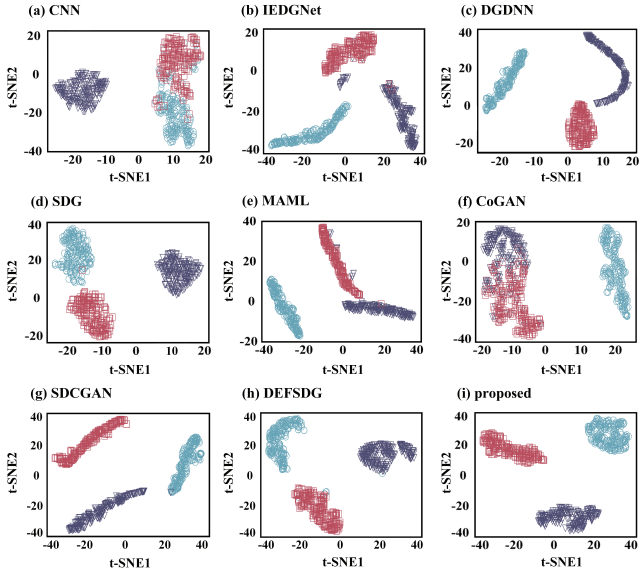

Fig. 20. Feature visualization of comparative methods on B1. (a) CNN. (b) IEDGNet. (c) DGDNN. (d) SDG. (e) MAML. (f) CoGAN. (g) SDCGAN. (h) DEFSDG. (i) Proposed.

To intuitively demonstrate the feature distribution of proposed method in unknown domain diagnosis tasks, t-SNE technology is introduced to visualize feature of the last hidden fully connected layer, illustrated in Figs. 18–20.

For the PDBN dataset, CNN, SDG, and IEDGNet fail to distinguish latent features of inner and outer ring faults. CoGAN shows almost no clear boundaries between the three health states. IEDGNet and DGDNN have distinct boundaries across health states but large intraclass distances. In contrast, the proposed method exhibits obvious decision boundaries with tight intraclass distances for all three states, showing that the simulation-enhanced MCCGAN and domain generalization network help build more generalizable and robust decision boundaries.

Feature visualization is also applied to the SDUST dataset to further highlight the proposed method's feature processing performance. CNN, SDG, and CoGAN have latent features that are nearly mixed and indistinct. IEDGNet and DGDNN show inconspicuous category boundaries. MAML, despite obvious boundaries, has confused features across categories. The proposed method, however, exhibits separable interclass and compact intraclass distances, strongly proving its outstanding capacity in unknown domain diagnosis tasks.

Feature visualization on the BJTU dataset further validates the proposed method's feature processing capabilities. CNN, SDG, and CoGAN have heavily overlapped, indistinct latent features with minimal class separability. IEDGNet and DGDNN show blurred class boundaries without clear clusters. MAML, although with apparent boundaries, has interclass confusion for certain faults. Notably, the proposed method yields features with clear interclass separability and compact intraclass clustering, showcasing superior discriminability. These results confirm its exceptional ability to learn discriminative representations, underpinning robustness in unknown domain tasks—especially in complex, real-world-like scenarios like the BJTU dataset.

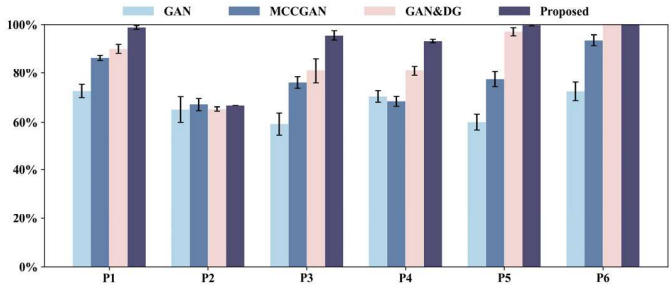

Fig. 21. Average accuracies of ablation methods on PDBN.

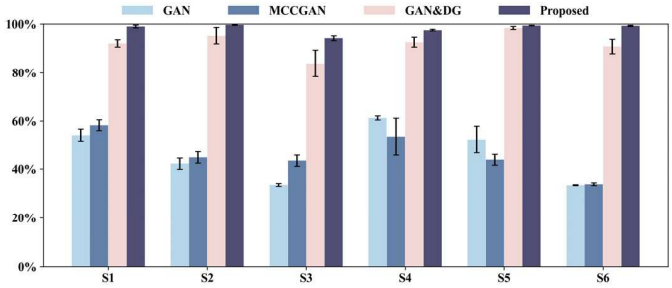

Fig. 22. Average accuracies of ablation methods on SDUST.

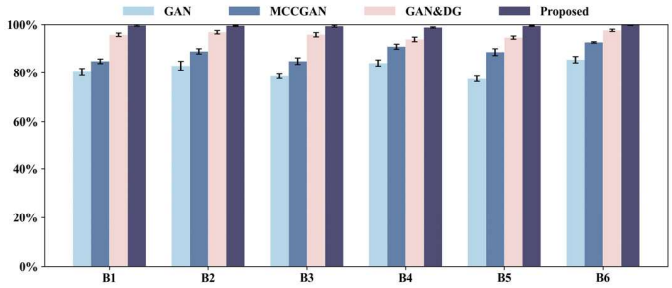

Fig. 23. Average accuracies of ablation methods on BJTU.

Across the PDBN, SDUST, and BJTU datasets, consistent patterns emerge: comparative methods struggle with indistinct feature separation, blurred boundaries, or large intraclass distances, while the proposed method consistently achieves clear interclass separability and compact intraclass clustering. This consistent superiority validates that the simulation-enhanced MCCGAN and domain generalization network enable robust, discriminative feature learning, underscoring the method's strong performance and generalizability in unknown domain diagnosis tasks—particularly in complex real-world scenarios.

E. Ablation Experiments

In this section, ablation experiments are conducted to evaluate the influence of the proposed MCCGAN and domain generalization network. As illustrated in Tables XI–XIII, Figs. 21–23, the diagnosis results of four ablation methods on 12 unknown domain diagnosis tasks are given.

On the PDBN dataset, without adding the domain generalization network, the average diagnostic accuracy of MCCGAN

TABLE XI
AVERAGE TESTING ACCURACIES (%) OF ABALATION METHODS ON PDBN

Dataset	Task	Metric	GAN	MCCGAN	GAN&DG	Proposed
PDBN	P1	*Acc*	72.53±2.72	86.40±1.08	90.13±1.81	**98.80±0.78**
		Precision	72.56±2.84	86.51±1.32	90.21±1.58	**98.85±0.75**
		Recall	72.49±2.51	86.33±0.94	90.25±1.49	**98.79±0.68**
		F1 Score	72.08±2.44	86.01±1.25	89.67±1.31	**98.21±0.87**
	P2	*Acc*	65.07±5.19	**67.07±2.41**	65.33±0.84	66.67±0.00
		Precision	65.01±5.08	**67.94±2.33**	65.29±0.79	66.67±0.00
		Recall	65.12±5.51	**67.12±2.77**	65.37±0.92	66.67±0.00
		F1 Score	64.73±4.77	**66.64±2.12**	65.08±0.79	66.67±0.00
	P3	*Acc*	59.20±4.45	76.00±2.35	80.93±5.12	**95.60±1.82**
		Precision	59.38±4.52	76.08±2.22	80.77±4.81	**95.62±1.74**
		Recall	59.23±4.19	75.94±2.42	80.84±4.95	**95.71±1.92**
		F1 Score	58.74±4.77	75.58±2.13	80.47±4.76	**95.34±1.77**
	P4	*Acc*	70.33±2.37	68.33±1.98	87.87±2.00	**93.33±0.73**
		Precision	70.41±2.18	68.28±1.87	87.76±1.96	**93.37±0.91**
		Recall	70.28±2.22	68.41±2.05	87.91±2.15	**93.29±0.87**
		F1 Score	69.97±2.12	67.92±1.87	87.44±1.84	**93.05±0.89**
	P5	*Acc*	60.00±3.24	77.33±3.07	97.07±1.61	**99.73±0.33**
		Precision	60.08±3.34	77.38±2.98	97.11±1.54	**99.78±0.43**
		Recall	59.89±3.05	77.34±3.02	97.12±1.75	**99.69±0.48**
		F1 Score	59.48±2.95	77.09±3.12	96.85±1.52	**99.31±0.45**
	P6	*Acc*	72.40±3.81	93.60±2.22	100.00±0.00	**100.00±0.00**
		Precision	72.41±3.88	93.64±2.38	100.00±0.00	**100.00±0.00**
		Recall	72.35±3.76	93.53±2.11	100.00±0.00	**100.00±0.00**
		F1 Score	72.24±3.59	93.21±2.05	100.00±0.00	**100.00±0.00**

TABLE XII
AVERAGE TESTING ACCURACIES (%) OF ABALATION METHODS ON SDUST

Dataset	Task	Metric	GAN	MCCGAN	GAN&DG	Proposed
SDUST	S1	*Acc*	54.09±2.46	58.19±2.25	92.07±1.50	**99.00±0.57**
		Precision	54.01±2.38	58.24±2.08	92.12±1.28	**99.12±0.49**
		Recall	54.12±2.11	58.14±2.31	92.04±1.59	**99.08±0.61**
		F1 Score	53.74±2.24	58.04±2.18	91.76±1.34	**98.73±0.49**
	S2	*Acc*	42.20±2.33	44.77±2.40	95.20±3.35	**99.60±0.14**
		Precision	42.23±2.34	44.81±2.39	95.23±3.41	**99.62±0.21**
		Recall	42.18±2.41	44.66±2.45	95.16±3.29	**99.57±0.18**
		F1 Score	41.85±2.18	44.59±2.26	94.89±3.16	**99.19±0.16**
	S3	*Acc*	33.50±0.57	43.42±2.30	83.87±5.50	**94.20±0.95**
		Precision	33.52±0.53	43.37±2.28	83.88±5.48	**94.26±0.81**
		Recall	33.55±0.49	43.38±2.35	83.91±5.61	**94.13±0.87**
		F1 Score	33.17±0.64	43.14±2.18	83.55±5.15	**93.84±0.95**
	S4	*Acc*	61.23±0.87	53.45±7.71	92.57±2.02	**97.50±0.32**
		Precision	61.24±0.91	53.51±7.38	92.64±2.17	**97.48±0.38**
		Recall	61.18±0.75	53.38±7.46	92.61±2.13	**97.39±0.45**
		F1 Score	61.04±0.91	53.19±7.67	92.24±2.06	**97.17±0.44**
	S5	*Acc*	52.23±5.58	43.80±2.20	98.37±0.58	**99.37±0.12**
		Precision	52.31±5.54	43.76±2.18	98.31±0.55	**99.39±0.14**
		Recall	52.28±5.61	43.84±2.34	98.29±0.62	**99.32±0.19**
		F1 Score	51.89±5.43	43.34±2.18	98.09±0.67	**99.04±0.18**
	S6	*Acc*	33.43±0.17	33.80±0.53	90.83±2.98	**99.24±0.23**
		Precision	33.47±0.21	33.78±0.49	90.76±3.02	**99.28±0.29**
		Recall	33.39±0.15	33.74±0.61	90.79±3.06	**99.23±0.28**
		F1 Score	33.17±0.20	33.69±0.58	90.42±2.48	**98.95±0.31**

is higher than that of GAN, but both methods are lower than the available accuracy. After adding the domain generalization network, the average diagnostic accuracy of MCCGAN and GAN is over 90%, but MCCGAN can improve the average diagnostic accuracy by 5% compared with GAN. This result demonstrates that the proposed MCCGAN can generate preferable augmented data than traditional GANs. The adversarial-based domain generalization network is able

TABLE XIII
AVERAGE TESTING ACCURACIES (%) OF ABALATION METHODS ON BJTU

Dataset	Task	Metric	GAN	MCCGAN	GAN&DG	Proposed
BJTU	B1	*Acc*	80.30±1.50	84.76±0.95	95.80±0.73	**99.63±0.29**
		Precision	80.18±1.34	84.70±0.84	95.74±0.52	**99.54±0.31**
		Recall	80.30±1.65	84.81±0.89	95.86±0.61	**99.70±0.34**
		F1 Score	79.70±1.37	84.15±0.82	95.28±0.58	**99.19±0.25**
	B2	*Acc*	82.90±1.85	88.90±1.10	96.85±0.68	**99.50±0.23**
		Precision	82.78±1.59	88.85±1.10	96.88±0.62	**99.45±0.26**
		Recall	82.85±1.96	89.00±1.05	96.98±0.64	**98.52±0.33**
		F1 Score	82.35±1.62	88.43±1.15	96.42±0.68	**98.97±0.32**
	B3	*Acc*	78.50±0.88	84.82±1.38	95.82±0.93	**99.37±0.36**
		Precision	78.18±0.79	84.75±1.34	95.90±0.78	**99.39±0.42**
		Recall	78.45±0.84	84.90±1.37	95.93±0.90	**99.43±0.32**
		F1 Score	77.85±0.92	84.38±1.39	95.50±0.81	**98.84±0.41**
	B4	*Acc*	84.05±1.26	90.87±1.03	93.87±0.93	**98.83±0.24**
		Precision	83.95±1.17	90.78±0.89	93.68±0.64	**98.76±0.30**
		Recall	84.07±1.18	90.95±0.98	93.75±0.88	**98.78±0.22**
		F1 Score	83.34±1.29	90.47±1.02	93.39±0.79	**98.48±0.24**
	B5	*Acc*	77.45±1.13	88.57±1.43	94.60±0.68	**99.43±0.17**
		Precision	77.05±0.98	88.53±1.14	94.45±0.44	**99.41±0.25**
		Recall	77.42±1.09	88.75±1.33	94.70±0.63	**99.48±0.15**
		F1 Score	76.88±1.04	88.37±1.22	94.10±0.57	**99.07±0.21**
	B6	*Acc*	85.43±1.35	92.63±0.29	97.63±0.48	**99.80±0.19**
		Precision	85.47±1.15	92.68±0.32	97.58±0.46	**99.84±0.22**
		Recall	85.50±1.26	92.45±0.28	97.66±0.42	**99.82±0.24**
		F1 Score	84.89±1.28	92.08±0.33	97.14±0.45	**99.45±0.21**

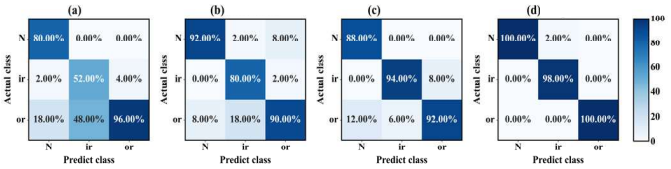

Fig. 24. Confusion matrix of ablation methods on P1. (a) GAN. (b) MCCGAN. (c) GAN&DG. (d) Proposed.

to enhance the generalization capacity of the diagnosis model in unknown domain diagnosis tasks.

Similar experiment results can be found in the SDUST dataset, as shown in Table XII and Fig. 22. This further validates the effectiveness of the proposed MCCGAN and domain generalization network. The proposed method is of great inspiring significance for unknown domain diagnosis tasks.

In addition, we validated the performance of our method on the BJTU high-speed train bogie dataset, a dataset closer to real-world scenarios, with diagnostic results demonstrated in Table XIII and Fig. 23. Ablation studies further confirmed that the proposed method holds significant potential for unknown domain diagnosis in real-world-like scenarios.

To further demonstrate the effectiveness of the two proposed modules, a confusion matrix is applied to analyze the category-wise results, as shown in Fig. 24. It can be observed that GAN and MCCGAN have great misdiagnosis rates in three health states. With the domain generalization network, the GAN method has a slight confusion in the classification of IR fault. In addition, the proposed method performs well in each health state, suggesting excellent data augmentation performance in the unknown target domain.

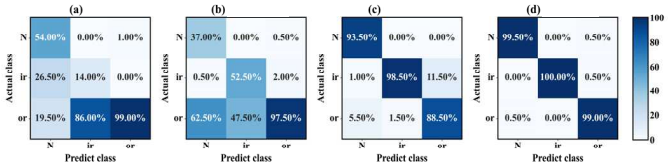

Fig. 25. Confusion matrix of ablation methods on S1. (a) GAN. (b) MCCGAN. (c) GAN&DG. (d) Proposed.

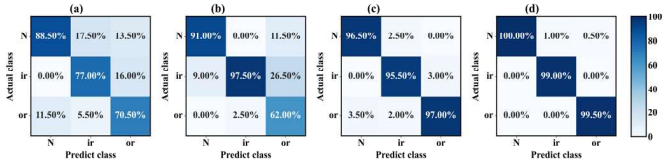

Fig. 26. Confusion matrix of ablation methods on B1. (a) GAN. (b) MCCGAN. (c) GAN&DG. (d) Proposed.

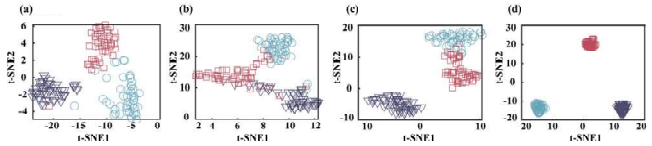

Fig. 27. Feature visualization of ablation methods on P1. (a) GAN. (b) MCCGAN. (c) GAN&DG. (d) Proposed.

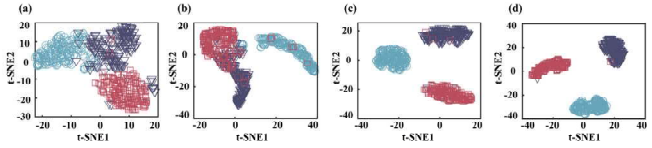

Fig. 28. Feature visualization of ablation methods on S1. (a) GAN. (b) MCCGAN. (c) GAN&DG. (d) Proposed.

On the SDUST dataset, similar results can also be gained. As illustrated in Fig. 25, although GAN and MCCGAN can distinguish the OR fault samples, the normal samples and IR fault samples can not be clarified well. With the help of domain generalization, the GAN method is able to clearly distinguish most samples, but still a small number of samples are in confusion. Analogously, the proposed method has almost no misdiagnosis in unknown domain diagnosis tasks.

On the BJTU dataset, comparable results are observed. As illustrated in Fig. 26, while GAN and MCCGAN can discriminate OR fault samples, normal samples, and IR fault samples are not well-distinguished. Through domain generalization, the GAN method clearly distinguishes most samples, although a small number still show confusion. Similarly, the proposed method achieves nearly error-free performance in unknown domain diagnosis tasks on this dataset.

To intuitively exhibit the feature distribution of ablation methods in unknown domain diagnosis tasks, t-SNE technology is employed to visualize features of the last hidden fully connected layer, as illustrated in Fig. 27 (PDBN), Fig. 28 (SDUST), and Fig. 29 (BJTU). It can be seen that the latent features of GAN and MCCGAN are confused together, and the latent features of GAN with the domain generalization method

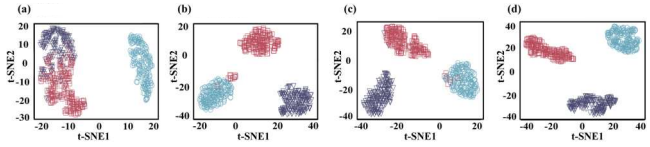

Fig. 29. Feature visualization of ablation methods on B1. (a) GAN. (b) MCCGAN. (c) GAN&DG. (d) Proposed.

were confused in normal and OR fault samples. In addition, it can be seen that the deep features of the proposed method in three health states have an obvious decision boundary with a tight interclass distance.

In addition, the visualized features are also demonstrated in Fig. 28. It can be observed that the deep features of the GAN and MCCGAN method has no category boundary to distinguish the three health states. With the help of domain generalization, the GAN method generates a distinct decision boundary, but the intraclass is not compact enough. In the proposed method, the deep features exhibit a separable interclass and compact intraclass distance. Consequently, these results strongly proved that the two proposed modules are effective in unknown domain diagnosis tasks.

In addition, visual features are also demonstrated in Fig. 29 for the BJTU dataset. It can be observed that the deep features of the GAN and MCCGAN methods lack clear category boundaries for distinguishing the three health states. Specifically, with the integration of domain generalization, the GAN method generates a distinct decision boundary; however, the intraclass features remain insufficiently compact. In contrast, the proposed method exhibits deep features with separable interclass distances and compact intraclass clusters. These results strongly demonstrate that the two proposed modules are effective in unknown domain diagnosis tasks, particularly in enhancing feature discriminability for complex industrial datasets like BJTU.

IV. Conclusion

This study addresses a challenging yet practical industrial scenario involving fault diagnosis in an unknown domain, where only a single known source domain is available for training. To overcome this challenge, a simulation-enhanced MCCGAN framework is proposed to facilitate deployment under previously unseen operating conditions. Specifically, a physical vibration model of bearings is developed to augment data from the unknown domain based on underlying mechanical principles. The MCCGAN, integrated with this simulation-augmented data, is employed to generate pseudo-target domain samples. Furthermore, an adversarial domain generalization network is introduced to improve the diagnostic model's generalization capability and robustness.

Comprehensive experiments conducted across three experimental platforms, benchmarked against state-of-the-art methods, demonstrate the superior generalization performance of the proposed approach in unknown domain diagnosis tasks. Notably, the proposed method achieves an average increase of 30% in test accuracy compared to a conventional CNN baseline, and surpasses other comparative methods by more

than 10% on average in diagnostic accuracy. In addition, the critical roles of both the MCCGAN and the domain generalization network are empirically validated.

A primary limitation of this work lies in the assumption that the operating condition information of the unknown domain is known a priori. Moreover, the reliance on simulated bearing vibration data may constrain the model's applicability to systems characterized by complex mathematical formulations or fault mechanisms, due to inherent assumptions regarding data distribution. Specifically, while the 4-DOF physical vibration model captures dominant fault dynamics, it cannot fully replicate the intricate interplay of multisource noise, progressive wear evolution, or unforeseen fault coupling effects present in real industrial machinery. Furthermore, fault degradation mechanisms can vary significantly across domains due to differences in operating conditions, loading profiles, and environmental factors. As a result, the fault characteristics and domain-specific information learned by a simulation or data generation model trained on a single source domain may not fully align with those present in unseen target domains. This discrepancy poses a fundamental challenge to the generalization of data-augmented diagnostic methods. Consequently, the model may still implicitly learn simulation-specific artifacts, and its generalization performance could degrade under extreme distribution shifts or in systems with poorly understood physical behaviors. Future research will focus on incorporating additional physical information and domain knowledge into the network construction process, with the aim of generating augmented data that more accurately reflects real-world attributes.

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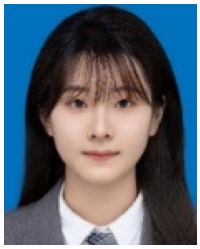

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